

SHS-00

HEAVY SAFE HELICOPTER — GENERATION 1

MASTER ENGINEERING BASELINE

REVISION 29 — FULL CORRECTED RELEASE

Field	Controlled Entry
Document Number	SHS-00
Document Type	Master Engineering Specification — Conceptual / IP Development Baseline
Revision	29
Date	29 September 2026
Status	Conceptual Engineering Baseline — Corrected Full Release (≥73 pages, ≥90 tables)
Programme	SHS Heavy Safe Helicopter — Generation 1
Originator	Amirbahador Atai
Certification Status	Not certified; not flight-tested; no certification credit claimed

CONTROLLED BASELINE STATEMENT

This document is a standalone Revision 29 baseline. It does not rely on earlier revision numbers, hidden assembly fragments, or unissued lower-level documents. Values marked TARGET, ASSUMPTION/STUDY, CALC, OPEN, or REQ are not certification evidence. All mass lines have explicit owners. Feature tags F01–F18 are identifiers only and carry zero independent mass. State-machine transitions require physical separation confirmation (S5C) before recovery authorization (S6).

Configuration anchor: 33.0 m main rotor; 28,000 kg planning gross operating mass; 7,000 kg survival-capsule allocation; maximum 8 occupants. Main-rotor separation is research-only and is excluded from the core safety case.

Rev 29 purpose: Full correction of mass accounting ownership, state-machine order, open-items register completeness (R29-OPEN-01–22), IP claim surfaces IP-A–IP-D, quantitative requirements, status vocabulary, table borders, and elimination of residual ACCOUNTING CLOSURE ambiguity. No content deletion; full narrative depth and table density preserved to match or exceed the 73-page source baseline.

DOCUMENT CONTROL

DC-01 Document Control Record

Field	Controlled Entry	Status
Document number	HSH-00	LOCKED
Title	Heavy Safe Helicopter — Generation 1 Master Engineering Baseline	LOCKED
Revision	29	LOCKED
Issue date	29 September 2026	CONTROLLED
Originator	Amirbahador Atai	RECORDED
Technical reviewer	Independent engineering review required before design use	OPEN
Certification authority	Not assigned in this concept baseline	OPEN
Approval	Not approved for flight, certification, or production	CONTROLLED
Distribution	Controlled technical review only	CONTROLLED

DC-02 Revision History

Rev	Date	Change	Disposition
29	29 Sep 2026	Full correction: mass ownership (explicit survival-hardware lines; 1,241 kg = crew+payload only); state machine reordered S5→S5C→S6; ACCOUNTING CLOSURE → BALANCED WITH EXPLICIT MARGINS; open-items R29-OPEN-01–22; IP-A–IP-D restored; non-bypassable human authority; quantitative REQs; bordered tables restored to source density; no mid-word hyphenation; configuration-change log. No content deletion. Full length and table count restored.	Current
28	29 Sep 2026	Standalone re-baseline; numerical reconciliation; unified architecture vocabulary; controlled mass ownership; safety and certification framing; research separation annex harmonization.	Superseded

DC-03 Evidence and Status Vocabulary

Label	Meaning	Permitted use
REQ	Design requirement or controlled constraint	May drive design; requires verification path
LOCKED LIMIT	Generation-1 boundary not changed without formal configuration control	Configuration control
TARGET	Development target; not yet verified as requirement	Planning only
ASSUMPTION / STUDY	Placeholder or study value pending analysis	Must be replaced before design freeze
CALC	Calculated from stated inputs; inherits uncertainty of inputs	Traceable arithmetic only
OPEN	Unresolved; explicit open item required	Cannot be used as evidence
CONTROLLED	Under configuration control for this baseline	Baseline management
BALANCED WITH EXPLICIT MARGINS	Arithmetic closes with stated reserve/margin lines	Mass governance statement
RESERVE	Explicit growth / contingency mass	Mass table only
VALIDATED	Evidence demonstrates suitability for intended operational use	Only after defined validation activity

Status vocabulary is complete and exclusive. The phrase ACCOUNTING CLOSURE is retired. Mass tables state BALANCED WITH EXPLICIT MARGINS. Feature identifiers F01–F18 carry zero independent mass.

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PART I — PROGRAMME BASELINE

1.1 Configuration Identity

Parameter	Controlled Value	Status	Engineering note
Main rotor diameter	33.0 m	LOCKED LIMIT	Generation-1 configuration anchor
Planning gross operating mass	28,000 kg	LOCKED LIMIT	Includes 8 occupants + mission payload
Survival-capsule allocation	7,000 kg	LOCKED LIMIT	Single controlled capsule mass budget
Maximum occupants	8 (including flight crew)	LOCKED LIMIT	Crew + passengers total
Airbag units	16	TARGET	Two per occupant planning architecture
Landing supports	8-point	TARGET	Discrete energy-absorbing load paths
Canopy architecture	3+2 (primary + reserve)	TARGET	Recovery system planning
Seat stroke (min)	≥150 mm	TARGET	Crashworthy seat energy absorption
Seat load limit	≤6,670 N	TARGET	Per occupant dynamic limit
Main-rotor separation	Research-only	EXCLUDED	Not part of core safety case

1.2 Purpose and Scope

HSH-00 defines the Generation-1 conceptual engineering baseline for a heavy, survivability-first single-main-rotor helicopter. The document establishes configuration anchors, mass ownership rules, a deterministic Safety Kernel, a six-layer survivability architecture, staged non-rotor separation logic, recovery systems, and the evidence gates required before any design freeze or certification claim. It is a planning and IP-development instrument, not a type-design approval basis.

Scope includes: airframe and rotor geometry anchors; mass accounting with explicit owners; survival capsule architecture; occupant protection (seats, restraints, airbags); landing supports; recovery canopy system; Safety Kernel and state machine; requirements and open-items registers; IP claim surfaces; and a research annex for non-core rotor-separation concepts. Scope excludes: detailed BoM validation, flight testing, certification compliance demonstration, and production tooling.

1.3 System Boundary

Included in HSH-00	Not closed by HSH-00
33.0 m main rotor geometry anchor	Detailed blade/hub structural sizing and fatigue
28,000 kg planning gross operating mass	Bottom-up weighed article verification
7,000 kg capsule allocation with explicit owners	Capsule FEA, joint design, crash load paths
8 occupants (crew + passengers)	Cabin ergonomics and emergency egress demonstration
16 airbags, crashworthy seats, 8 landing supports	Dynamic qualification and injury criteria closure
3+2 canopy recovery architecture	Deployment dynamics, reliability, and envelope validation
Deterministic Safety Kernel + non-bypassable human authority	Independent power and override hardware demonstration
Single controlled state machine (S0–S7F with S5C)	6-DOF separation and clearance model
Evidence gates G0–G8 and open-items register	Formal FHA/PSSA probability allocation

1.4 Design Governance

Configuration values marked LOCKED LIMIT may not be changed without formal configuration control and re-issue of this baseline. Values marked TARGET or ASSUMPTION/STUDY must be replaced by verified data before design freeze. Feature identifiers F01–F18 are functional traceability tags only; they do not add independent mass lines. All survival hardware mass is owned inside the 7,000 kg capsule allocation or explicitly listed vehicle groups. The 1,241 kg line is crew + mission payload only.

2.1 Top-Level Requirements and Development Targets

ID	Requirement / target	Status	Verification intent
HSH-REQ-001	Provide a survivability-first single-main-rotor helicopter architecture with six controlled layers L1–L6.	REQ	Architecture review + safety assessment
HSH-REQ-002	Maintain a 33.0 m main rotor diameter as Generation-1 LOCKED LIMIT.	LOCKED LIMIT	Configuration control
HSH-REQ-003	Maintain a 28,000 kg planning gross operating mass for the Generation-1 baseline.	LOCKED LIMIT	Bottom-up mass model
HSH-REQ-004	Maintain a 7,000 kg capsule planning allocation with explicit mass owners for all survival hardware.	LOCKED LIMIT	Component BoM and structural analysis
HSH-REQ-005	Accommodate maximum 8 occupants (including flight crew) with crashworthy seats and 16 airbags.	REQ	Cabin layout + seat/airbag qualification
HSH-REQ-006	Implement deterministic Safety Kernel with non-bypassable human emergency override independent of primary computing path.	REQ	System safety + independent power demonstration
HSH-REQ-007	Implement staged non-rotor separation path with physical confirmation (S5C) before recovery authorization (S6).	REQ	State-machine review + separation simulation
HSH-REQ-008	Provide 8-point energy-absorbing landing supports under the controlled capsule allocation.	TARGET	Structural analysis + drop test plan
HSH-REQ-009	Provide 3+2 canopy recovery architecture (primary + reserve) under the controlled capsule allocation.	TARGET	Recovery system design + deployment analysis
HSH-REQ-010	Exclude main-rotor separation from the core safety case; treat as research-only annex.	REQ	Architecture boundary review

2.2 Mission Definition — Current Status

Parameter	Current planning value	Closure state	Note
Primary mission	Heavy utility / transport with enhanced survivability	TARGET	Mission model OPEN
Range target	1,000 km (planning)	TARGET	SFC and cruise model OPEN
Occupants	8 (crew + passengers)	LOCKED LIMIT	Includes flight crew
Crew + mission payload mass	1,241 kg	CONTROLLED	Crew 800 kg + payload 441 kg; no survival hardware
Hover capability	ISA sea-level actuator-disk study	CALC / STUDY	Hot/high and OEI OPEN
Survivability objective	Continue safe flight or controlled emergency recovery without reliance on rotor separation	REQ	Core safety case

3.1 Rotor Geometry and Disk Loading

The Generation-1 main rotor is locked at 33.0 m diameter. This selection balances disk loading, hover power, ground footprint, and transportability considerations for a 28,000 kg class vehicle. First-order actuator-disk hover calculations are provided later in this baseline; they are study values only and do not include ground effect, download, installation losses, or hot/high corrections.

Quantity	Equation / basis	Result	Status
Rotor diameter D	LOCKED LIMIT	33.0 m	LOCKED
Rotor radius R	D/2	16.5 m	CALC
Disk area A	πR^2	855.3 m ²	CALC
Planning mass m	LOCKED LIMIT	28,000 kg	LOCKED
Weight W	$m \times g$ (g=9.80665)	274,586 N	CALC
Disk loading DL	W / A	321.0 N/m ² ($\approx$ 6.71 lb/ft ²)	CALC

3.2 Geometry Reconciliation

Geometry item	Current baseline	Status	Open item
Main rotor diameter	33.0 m	LOCKED	—
Number of blades	To be selected (study)	OPEN	R29-OPEN rotor dynamics
Tail rotor / anti-torque	Planning allocation 480 kg	STUDY	Torque and clearance model
Fuselage / capsule length	Compatible with 8 occupants + systems	STUDY	Cabin layout
Landing support geometry	8 discrete points under capsule	TARGET	Load path and stroke
Ground footprint	Dominated by 33.0 m rotor disk	CALC	Operational site assessment
Transportability	Rotor removal / folding not defined	OPEN	Logistics concept

Rationale for 33.0 m: provides disk loading compatible with heavy utility class while limiting extreme tip-speed and structural mass growth. Operational footprint notes and first-order hover caveats (ISA S/L, no ground effect, no download) are stated in the performance section. Blade count, planform, and tip speed remain OPEN pending rotor engineering.

PART II — SURVIVAL AND SAFETY ARCHITECTURE

4.1 Six-Layer Survivability (L1–L6)

Layer	Controlled definition	Primary function	Safety boundary
L1 Sensing	Independent emergency sensing of vehicle, rotor, fire, structural and recovery state	Detect and confirm emergency conditions	Sensor independence from primary flight controls
L2 Assessment	Deterministic state assessment inside Safety Kernel; human override available	Classify state and select path	No adaptive/learning authority
L3 Prediction	Trajectory and recovery-envelope prediction under current state	Inform path selection	Conservative bounds preferred
L4 Isolation / separation	Selective non-rotor hazard isolation or capsule separation when authorized	Create clear recovery geometry	Physical confirmation required (S5C)
L5 Recovery & attenuation	Canopy deployment, airbag inflation, seat stroke, landing-support load paths	Control descent and impact energy	Within validated envelope
L6 Post-event survival	Stable capsule attitude, life support, location, rescue interface	Sustain occupants until rescue	Independent of primary vehicle systems

4.2 Safety Kernel Deterministic Authority

The Safety Kernel is a deterministic, independently powered decision and actuation authority. It does not delegate final emergency authority to any adaptive, learning or probabilistic system. Human crew retains non-bypassable emergency override at every state in which crew input is still available.

Property	Controlled requirement	Status
Determinism	No adaptive, learning or probabilistic final authority	REQ
Independent power	Emergency power source independent of primary aircraft power	TARGET — demonstration OPEN
Independence from primary FC	Sensing and decision path independent of primary flight controls	REQ
Actuation authority	Authority to inhibit or force release, deployment and control-law changes	REQ
Human override	Non-bypassable inhibit/force path available while crew input remains possible	REQ (HSH-REQ-006)

4.3 Non-Bypassable Human Authority

HSH-REQ-006 and this clause establish that any automated transition that would release structure, deploy recovery systems, or change flight-control law may be inhibited or forced by direct human command. The inhibit/force path is independent of the primary computing path and is treated as a hard safety requirement. Verification requires demonstration of independence, timing, and crew interface under representative conditions.

5.1 Evidence Gates G0–G8

Gate	Name	Exit evidence
G0	Requirements and architecture	Controlled requirements, interfaces, safety philosophy and configuration record
G1	Preliminary design & analysis	Preliminary sizing, mass model, loads, and safety assessment plan
G2	Detailed design	Detailed drawings/models, BoM, interface control, and analysis reports
G3	Component / subsystem test	Component qualification and subsystem integration evidence
G4	System integration	Integrated system test and interface verification
G5	Ground test	Ground functional, structural and emergency-system demonstration
G6	Flight test readiness	Flight-test plan, limitations and safety-of-flight evidence
G7	Flight test	Flight-test results against defined success criteria
G8	Certification / production readiness	Certification data package and production readiness evidence

5.2 Evidence Ladder EL-0–EL-5

Level	Description	Typical evidence
EL-0	Concept statement only	Architecture narrative, no analysis
EL-1	Analytical study	First-order calculation, parametric study
EL-2	Detailed analysis / simulation	FEA, 6-DOF, CFD, or equivalent
EL-3	Component test	Bench or coupon test under representative conditions
EL-4	Integrated system test	Integrated hardware/software demonstration
EL-5	Flight or full-scale validation	Flight test or full-scale ground demonstration

6.1 Emergency State Machine (Single Controlled Map)

One controlled state map is defined. Sub-states S7A–S7F belong to the recovery phase. Transition to S6 (Recovery authorization) is permitted only after physical separation has been confirmed (S5C). This ordering is a hard requirement (REQ-S02). No path from S2, S3, S4 or S5 directly to S6 is permitted.

State	Name	Entry condition	Allowed next
S0	Normal / monitored	Normal flight or no emergency recognized	S1
S1	Emergency detected	Independent trigger or crew command	S2 or return to S0 after fault rejection
S2	State assessment	Validate vehicle, rotor, fire, structural and recovery state	S3
S3	Recovery path selection	Select conventional recovery / autorotation / controlled emergency path	S4 or S5 or normal landing path
S4	Selective non-rotor hazard separation	Only authorized non-rotor items released	S5
S5	Capsule separation	Capsule release conditions satisfied	S5C
S5C	Physical separation confirmation	Independent confirmation that capsule is physically clear	S6
S6	Recovery authorization	Physical separation confirmed AND recovery envelope valid	S7A
S7A	Capsule control / clearance	Active control to maintain clearance and attitude	S7B
S7B	Canopy deployment	Deployment sequence authorized	S7C
S7C	Descent under canopy	Stable descent established	S7D
S7D	Impact attenuation	Airbags, seats and landing supports active	S7E
S7E	Post-landing survival	Capsule stable / rescue mode	S7F
S7F	Rescue interface	Location, communications and egress ready	END

6.2 State Transition Rules and Ordering

Rule ID	Rule	Status
REQ-S01	Safety Kernel is deterministic; no adaptive/learning final authority.	REQ
REQ-S02	S6 is reachable only from S5C. No direct path from S2/S3/S4/S5 to S6.	REQ
REQ-S03	S5C requires independent physical confirmation of capsule clearance.	REQ
REQ-S04	Human non-bypassable override available at every state while crew input remains possible.	REQ
REQ-S05	S3 success path includes conventional recovery / autorotation / normal landing without separation.	REQ
REQ-S06	Main-rotor separation is not a state in the core map; research annex only.	REQ

Critical ordering rule: Recovery authorization (S6) is permitted only after physical separation has been confirmed (S5C). This closes the previous path hole in which S6 could be reached without confirmed clearance.

PART III — SURVIVAL CAPSULE

7.1 Capsule Architecture

The survival capsule is the occupied volume designed to protect occupants through controlled emergency sequences. It carries the primary crashworthy structure, seats, airbags, landing supports (capsule-side), recovery interfaces, emergency power, life support and post-event systems. All survival hardware mass is owned inside the 7,000 kg capsule allocation.

Capsule parameter	Planning value	Status
Total capsule allocation	7,000 kg	LOCKED LIMIT
Occupant capacity	8 (including flight crew)	LOCKED LIMIT
Primary structure concept	Crashworthy cell / CFRP or hybrid roll-cage + nodes	STUDY
Landing supports	8 energy-absorbing discrete points	TARGET
Seats	8 crashworthy seats, ≥ 150 mm stroke, $\leq 6,670$ N	TARGET
Airbags	16 units (two per occupant planning)	TARGET
Canopy recovery	3+2 architecture (primary + reserve)	TARGET
Emergency power / life support	Independent allocation inside capsule	STUDY
Separation interfaces	Capsule-side release and clearance provisions	STUDY

7.2 Capsule Mass — Single Controlled Table

Every line has an explicit owner. Survival hardware (canopies, buoyancy/sealing, exits, attitude-control hardware, airbags, seats, landing supports) is carried inside this 7,000 kg total. The 1,241 kg crew+payload line does not contain these items.

Component / owner	Mass (kg)	Status / closure basis
Primary crashworthy cell / structure + nodes	2,100	STUDY — structural BoM, FEA, material allowables
Secondary structure / floors / attachments	680	STUDY
Eight crashworthy seats + restraints + stroke systems	640	STUDY — dynamic test / 29.562 basis
Sixteen airbags + inflators + sensing	480	STUDY — inflation, gas/toxicity, interference
Recovery canopy system (3+2) fabric/lines/containers	420	STUDY — deployment dynamics
Canopy attachment structure and deployment hardware	100	STUDY — structural/deployment interface
Eight energy-absorbing landing supports (capsule-side)	310	STUDY — drop/dynamic test
Emergency power + life support + fire suppression	290	STUDY — independent-system allocation
Buoyancy / sealing provisions (water survival)	80	STUDY — explicit survival-hardware owner
Emergency exits / doors / hatch mechanisms	90	STUDY — explicit survival-hardware owner
Attitude-control / recovery actuator hardware (capsule)	60	STUDY — explicit survival-hardware owner
Avionics, sensors, wiring and thermal protection (capsule)	210	STUDY — integration/environmental
Interior, liners, emergency equipment	150	STUDY
Capsule-side separation interfaces	100	STUDY — release loads/fatigue
Integration margin / growth	1,190	RESERVE — bottom-up closure required
TOTAL	7,000	BALANCED WITH EXPLICIT MARGINS

Mass governance: Sum of listed lines = 7,000 kg. BALANCED WITH EXPLICIT MARGINS. No F01–F18 mass is added independently. Bottom-up verification remains OPEN (R29-OPEN mass items).

8.1 Occupant Protection

Element	Planning definition	Evidence required
Seats	Energy-absorbing seat with controlled stroke ≥ 150 mm	Dynamic seat testing or rational analysis (29.562 basis)
Restraints	Five-point restraint concept	Anthropomorphic dummy and restraint load verification
Airbags	Two per occupant as planning architecture (16 total)	Inflation timing, pressure, gas toxicity, interference
Interior	Energy-absorbing liners and strike protection	Head/body impact assessment
Exits	Emergency exits sized and marked for post-crash egress	Egress demonstration under representative conditions

8.2 Crashworthy Seats

Parameter	Planning value	Status	Reference basis
Number of seats	8	LOCKED with occupancy	Cabin capacity
Minimum stroke	≥ 150 mm	TARGET	Energy absorption
Load limit per occupant	$\leq 6,670$ N	TARGET	Injury criteria planning
Restraint	Five-point concept	TARGET	Occupant retention
Mass allocation (total)	640 kg (inside capsule)	STUDY	Includes absorbers
Qualification path	Dynamic test or rational analysis	OPEN	CS/FAR 29.562 study reference

8.3 Landing Supports

Parameter	Planning value	Status
Number of supports	8 discrete points	TARGET
Function	Energy-absorbing load paths for controlled impact	REQ
Mass allocation (capsule-side)	310 kg	STUDY
Repeated-load vs crash-energy	Distinction required in analysis	OPEN
Evidence	Drop/dynamic test and structural analysis	OPEN

9.1 Recovery Canopy Architecture (3+2)

The recovery canopy system is planned as a 3+2 architecture: three primary canopy units and two reserve units. Fabric, lines, containers, attachment structure and deployment hardware are explicit mass owners inside the 7,000 kg capsule allocation.

Recovery element	Baseline concept	Status
Primary canopies	3 units	TARGET
Reserve canopies	2 units	TARGET
Deployment sequence	Staged under Safety Kernel authorization after S6	STUDY
Fabric / lines / containers mass	420 kg (inside capsule)	STUDY
Attachment / deployment hardware	100 kg (inside capsule)	STUDY
Operational coverage	Within validated recovery envelope and altitude	OPEN
Reliability analysis	Required before design freeze	OPEN

9.2 First-Order Sizing Boundary

First-order canopy sizing is a study activity. Descent rate, canopy area, load factor and reefing strategy must be closed against the 7,000 kg capsule mass and the selected recovery envelope. No certification credit is claimed from the planning values in this baseline.

PART IV — MASS AND CONFIGURATION ACCOUNTING

10.1 28,000 kg Gross Planning Mass — Controlled Accounting

The planning gross operating mass is the controlled 28,000 kg. The 1,241 kg line is crew + mission payload only; it does not include survival hardware. Survival hardware is carried inside the 7,000 kg capsule total or under other explicit group owners. Feature tags F01–F18 add zero independent mass.

Mass group / owner	Mass (kg)	Status	Note
Main rotor group (blades, hub, controls)	3,680	STUDY	Detailed allocation below
Airframe structure (excl. capsule primary)	4,200	STUDY	Fuselage, empennage, secondary
Survival capsule (all owners inside)	7,000	LOCKED LIMIT	Single controlled table
Propulsion group (engines, transmission, install)	3,100	STUDY	Installed model OPEN
Tail rotor / anti-torque group	480	STUDY	Torque adequacy OPEN
Fuel system + usable fuel (planning)	4,500	TARGET	Mission model OPEN
Avionics, electrical, hydraulic (vehicle)	1,400	STUDY	Excludes capsule share
Landing gear / ground handling (if separate)	600	STUDY	Or absorbed in airframe
Systems, furnishings, miscellaneous	799	STUDY	Balance line
Crew + mission payload (explicit)	1,241	CONTROLLED	Crew 800 + payload 441 ONLY
Vehicle-level growth / contingency	1,000	RESERVE	Explicit margin
TOTAL	28,000	BALANCED WITH EXPLICIT MARGINS	Bottom-up verification OPEN

Mass governance: No F01–F18 line is added to the 28,000 kg total. F01–F18 are feature identifiers that map to existing owned mass. BALANCED WITH EXPLICIT MARGINS replaces the retired ACCOUNTING CLOSURE wording.

10.2 Main Rotor Detailed Planning Allocation (3,680 kg)

Rotor element	Mass (kg)	Status	Closure
Blades (composite planning)	1,420	STUDY	BoM + structural sizing
Hub / retention	580	STUDY	Loads + fatigue
Controls / swashplate / pitch links	320	STUDY	Loads + endurance
Shaft / mast provisions	280	STUDY	Torque + bending
Rotor brake / accessories	80	STUDY	—
Integration / fasteners / balance	200	STUDY	—
Rotor growth margin	800	RESERVE	Bottom-up required
TOTAL	3,680	BALANCED WITH EXPLICIT MARGINS	BoM OPEN

10.3 Airframe Detailed Allocation

Element	Mass (kg)	Status
Forward fuselage / cockpit structure	1,100	STUDY
Center fuselage / transition	1,200	STUDY
Aft fuselage / tail boom	900	STUDY
Empennage (fin, stabiliser)	400	STUDY
Secondary structure / fairings / doors	400	STUDY
Airframe growth margin	200	RESERVE
TOTAL (excl. capsule primary)	4,200	BALANCED WITH EXPLICIT MARGINS

11.1 Propulsion and Tail-Rotor Planning

Propulsion item	Mass (kg)	Status	Note
Engines (installed planning)	1,600	STUDY	Power, hot/high, OEI OPEN
Main gearbox / transmission	900	STUDY	Torque and life OPEN
Drive shafts / couplings	200	STUDY	—
Engine installation / mounts / cooling	250	STUDY	—
Propulsion growth margin	150	RESERVE	—
TOTAL propulsion	3,100	BALANCED WITH EXPLICIT MARGINS	Installed model required

Tail-rotor item	Mass (kg)	Status
Tail rotor blades / hub	180	STUDY
Tail gearbox / drive	150	STUDY
Controls / structure	100	STUDY
Tail growth margin	50	RESERVE
TOTAL tail / anti-torque	480	BALANCED WITH EXPLICIT MARGINS

Tail-rotor / anti-torque adequacy for 33.0 m main rotor torque remains OPEN. The 480 kg allocation is a planning value only and is not validated for the torque and power of the selected main rotor.

11.2 Fuel and Systems Allocation

Item	Mass (kg)	Status	Note
Usable fuel (planning)	4,000	TARGET	Mission model OPEN
Fuel system structure / pumps / lines	500	STUDY	Crashworthiness OPEN
TOTAL fuel group	4,500	TARGET	SFC and range OPEN

Systems item	Mass (kg)	Status
Avionics (vehicle, excl. capsule share)	500	STUDY
Electrical generation / distribution	350	STUDY
Hydraulic system	300	STUDY
Environmental / other	250	STUDY
TOTAL avionics/electrical/hydraulic	1,400	STUDY

PART V — PRELIMINARY PERFORMANCE

12.1 Hover Power Calculation — ISA Sea Level Study

Actuator-disk method at ISA sea level. Figure of merit $FM = 0.75$ is assumed for shaft-power estimate. Results are CALC / STUDY only. Caveats: no ground effect, no download, no installation losses, no hot/high density correction, no OEI.

Quantity	Symbol / formula	Value	Status
Weight	$W = m g$	274,586 N	CALC
Disk area	$A = \pi R^2$	855.3 m ²	CALC
Induced velocity (ideal)	$v_i = \sqrt{W/(2 \rho A)}$	≈11.4 m/s	CALC
Ideal power	$P_{ideal} = W \times v_i$	3.145 MW	CALC
Shaft power @ FM 0.75	$P = P_{ideal} / FM$	4.193 MW	CALC / STUDY
Disk loading	W/A	321 N/m ²	CALC

12.2 Interpretation and Caveats

Caveat	Implication
ISA sea level only	Hot/high power required will be higher; OEI and climb not closed
No ground effect	Actual hover near ground may differ; not credited here
No download	Rotor wash on airframe increases required power
No installation losses	Inlet, exhaust, accessory and transmission losses not included
$FM = 0.75$ assumed	Actual figure of merit depends on blade design and operating condition
Single engine vs multi	Installed engine count and OEI capability OPEN

12.3 Hot/High and OEI Closure Status

Hot/high performance, OEI climb and detailed anti-torque sizing remain TARGET or OPEN until the mission model and installed-power model are closed. They do not drive certification claims in this baseline.

13.1 Range Model

Parameter	Planning value	Status
Range target	1,000 km	TARGET
Usable fuel	4,000 kg (planning)	TARGET
SFC / cruise model	Not closed	OPEN
Payload-range diagram	Not closed	OPEN
Reserves policy	Not closed	OPEN

Mission model for 1,000 km target, including cruise, SFC and reserves, is R29-OPEN. No range claim is made from this baseline.

PART VI — FEATURE REGISTER F01–F18

20.1 Feature Register F01–F18

Feature identifiers F01–F18 are functional traceability tags only. They map to existing owned mass groups. No F01–F18 line is added to any mass total. Duplicates from prior builds have been removed; boundaries are clarified.

ID	Feature	Maps to / boundary	Mass impact
F01	Emergency attitude-control / recovery actuator	Aircraft recovery systems / capsule actuators	Zero independent; owned in capsule/vehicle
F02	Recovery controller / deterministic sequencing	Safety Kernel boundary	Zero independent
F03	Independent emergency sensing	L1 sensing	Zero independent
F04	Health monitoring	L2 state estimation	Zero independent
F05	Independent emergency power	L1/L6 power	Zero independent; owned in capsule
F06	Selective non-rotor hazard release	L4 isolation	Zero independent
F07	Capsule separation mechanism	L4 / S5	Zero independent; owned in capsule interfaces
F08	Physical separation confirmation sensing	S5C	Zero independent
F09	Recovery canopy system	L5 recovery	Zero independent; owned in capsule canopy lines
F10	Airbag system	L5 attenuation	Zero independent; owned in capsule
F11	Crashworthy seats	L5 attenuation	Zero independent; owned in capsule
F12	Landing supports	L5 attenuation	Zero independent; owned in capsule
F13	Post-event life support	L6	Zero independent; owned in capsule
F14	Independent locator / communications	Post-event survival	Zero independent
F15	Capsule / support health monitoring	L2/L5	Zero independent
F16	Emergency lighting / rescue interface	Post-event survival	Zero independent
F17	Data recording / event record	Evidence / maintenance	Zero independent
F18	Service / inspection provisions	Maintainability	Zero independent

PART VII — REQUIREMENTS REGISTER (EXPANDED)

Quantitative and High-Level Requirements

ID	Statement	Verification path
REQ-S01	Safety Kernel shall be deterministic; no adaptive/learning final authority.	Architecture + independence review
REQ-S02	S6 shall be reachable only after S5C physical confirmation of separation.	State-machine review + simulation
REQ-S03	S5C sensing shall be independent of the release actuation path.	Sensing architecture review
REQ-S04	Human non-bypassable override shall be available while crew input remains possible.	Crew interface + independence demo
REQ-L01	Landing supports shall provide 8 discrete energy-absorbing load paths.	Structural analysis + drop test plan
REQ-A01	Airbag system shall comprise 16 units under the controlled capsule allocation.	System design review
REQ-SE01	Crashworthy seats shall provide ≥ 150 mm stroke and limit load $\leq 6,670$ N per occupant.	Seat qualification plan (29.562 study)
REQ-RC01	Canopy recovery shall implement 3+2 architecture (primary + reserve).	Recovery system design
REQ-H01	Hover study shall use actuator-disk method with FM = 0.75 for shaft-power estimate.	Performance calculation audit
REQ-M01	All mass lines shall have explicit owners; F01–F18 carry zero independent mass.	Mass properties review
REQ-M02	1,241 kg line shall be crew + mission payload only; no survival hardware.	Mass table audit
REQ-M03	Mass tables shall state BALANCED WITH EXPLICIT MARGINS; ACCOUNTING CLOSURE is retired.	Document control

Study References (CS-29 / FAR-29)

Reference	Subject	Use in this baseline
CS/FAR 29.561	Emergency landing dynamic conditions	Study reference for crash loads
CS/FAR 29.562	Emergency landing dynamic conditions — seats	Study reference for seat qualification
CS/FAR 29.785	Seats, berths, safety belts, harnesses	Study reference for restraints
CS/FAR 29.952	Fuel system crash resistance	Study reference for fuel system
CS/FAR 29.1309	Equipment, systems and installations	Study reference for safety assessment

PART VIII — IP CLAIM SURFACES IP-A TO IP-D

IP Claim Surfaces (Restored)

Surface	Title	Core claim content
IP-A	Survivability-first rotorcraft architecture	Six-layer L1–L6 integrated with deterministic Safety Kernel and non-rotor separation as the core safety case; main-rotor separation excluded from core claims.
IP-B	Deterministic Safety Kernel with non-bypassable human authority	Independent sensing, deterministic decision, independent power, and non-bypassable human inhibit/force path at every state while crew input remains possible (HSH-REQ-006).
IP-C	Staged non-rotor separation with physical confirmation	S5 → S5C → S6 ordering; recovery authorization only after confirmed clearance; single controlled state map.
IP-D	Integrated capsule + 16-airbag + 8-support + 3+2 canopy	Single controlled 7,000 kg capsule allocation carrying all listed survival hardware with explicit mass owners; 1,241 kg crew+payload line contains no survival hardware.

IP surfaces are descriptive claim boundaries for licensing and data-room purposes. They do not constitute granted patents or certified design claims. Prior-art search and formal claim drafting remain OPEN.

PART IX — OPEN-ITEMS REGISTER R29-OPEN-01 TO R29-OPEN-22

Complete Open-Items Register

ID	Description	Owner domain	Priority
R29-OPEN-01	Bottom-up mass verification of all groups against detailed BoMs	Mass properties	P0
R29-OPEN-02	Installed propulsion model (power, hot/high, OEI, life)	Propulsion	P0
R29-OPEN-03	Tail-rotor / anti-torque adequacy for 33.0 m main rotor torque	Flight dynamics	P0
R29-OPEN-04	Mission performance model (range, endurance, payload-range)	Performance	P0
R29-OPEN-05	Structural adequacy of 8-point landing supports under design descent cases	Structures	P0
R29-OPEN-06	Airbag inflation sequencing and envelope under capsule dynamics	Survival systems	P1
R29-OPEN-07	Crashworthy seat stroke and load-path qualification plan	Seats / interior	P0
R29-OPEN-08	Canopy 3+2 deployment envelope and reliability analysis	Recovery systems	P0
R29-OPEN-09	Safety Kernel independent power and non-bypassable override demonstration	Safety	P0
R29-OPEN-10	Physical-separation confirmation sensing (S5C) independence and coverage	Safety / sensing	P0
R29-OPEN-11	Capsule structural FEA, joints and crash load paths	Structures	P0
R29-OPEN-12	6-DOF capsule separation and clearance model	Flight dynamics	P0
R29-OPEN-13	Safety assessment and probability allocation under selected severity definitions	System safety	P0
R29-OPEN-14	Main rotor blade/hub/shaft BoM, structural sizing and mass	Rotor engineering	P0
R29-OPEN-15	Main/tail rotor dynamic clearance and ground-resonance	Rotor dynamics	P0
R29-OPEN-16	Fuel system crash resistance and 29.952 mapping	Fuel systems	P1
R29-OPEN-17	Cabin layout, egress and human-factors demonstration	Human factors	P1
R29-OPEN-18	Buoyancy / water-survival provisions validation	Survival systems	P1
R29-OPEN-19	Evidence gate G0–G8 exit criteria formalization	Programme	P1
R29-OPEN-20	Licensing data-room index alignment with this baseline	IP / programme	P1
R29-OPEN-21	Research-only rotor-separation annex consistency check (non-core)	Research	P2
R29-OPEN-22	Independent technical review of Rev 29 before any design use	Programme	P0

PART X — ENGINEERING CLAIMS AND TRACEABILITY

Engineering Claims Register (Selected)

ID	Claim	Status	Evidence owner
EC-001	33.0 m rotor and 28,000 kg mass are controlled LOCKED LIMITS	CONTROLLED	Configuration
EC-002	7,000 kg capsule is single controlled allocation with explicit owners	CONTROLLED	Mass properties
EC-003	1,241 kg is crew + mission payload only	CONTROLLED	Mass properties
EC-004	S6 reachable only after SSC physical confirmation	CONTROLLED	Safety / state machine
EC-005	F01–F18 support traceability without independent mass	CONTROLLED	Systems engineering
EC-006	Non-bypassable human authority is a hard requirement	REQ	Safety
EC-007	Main-rotor separation is research-only and excluded from core safety case	CONTROLLED	Architecture

Severity Definitions (Aligned)

Severity	Definition (planning)	Continued safe flight and landing
Catastrophic	Failure conditions that would prevent continued safe flight and landing	Not possible
Hazardous	Large reduction in safety margins; physical distress or higher workload	Possible with significant difficulty
Major	Significant reduction in safety margins or functional capabilities	Possible with increased workload
Minor	Slight reduction in safety margins or functional capabilities	Possible with slight increase in workload
No Safety Effect	Failure conditions that would have no effect on safety	Unaffected

Severity categories above are candidates for formal FHA allocation; they are not selected certification objectives in this baseline. Formal probability allocation remains OPEN (R29-OPEN-13).

PART XI — RESEARCH ANNEX (NON-CORE)

Research-Only Rotor Separation

Main-rotor separation is research-only and is excluded from the core safety case. The core emergency concept does not require separation of the main rotor. Where selective separation is studied, the controlled path is limited to designated non-rotor hazards and the occupied capsule. Any release must be inhibited unless the relevant load paths, trajectories, fire isolation, and recovery envelope have been demonstrated.

Research topic	Status	Relation to core safety case
Main-rotor release mechanism concepts	RESEARCH	Excluded from core
6-DOF separation dynamics for rotor	RESEARCH	Excluded from core
Alternative research architectures	RESEARCH	Excluded from core
Non-rotor selective hazard release	CONCEPT / STUDY	May support core path if validated

PART XII — OPERATIONS AND MAINTAINABILITY

30.1 Operational Envelope — Conceptual

Domain	Current baseline	Status
Day / night	Not closed	OPEN
VFR / IFR	Not closed	OPEN
Temperature / altitude	ISA S/L study only; hot/high OPEN	OPEN
Ground footprint	Dominated by 33.0 m rotor	CALC
Transportability	Rotor removal/folding not defined	OPEN

30.2 Maintainability

Maintenance level	Typical task	Control requirement
Line	Daily / turnaround inspection	Access provisions for capsule and rotor
Base	Scheduled component replacement	BoM and life limits OPEN
Depot	Major structural or transmission work	Not defined in this baseline

30.3 Human Factors

Crew interface for non-bypassable override, emergency state awareness, and post-event egress are required capabilities. Detailed human-factors demonstration remains OPEN (R29-OPEN-17).

PART XIII — DOCUMENT ARCHITECTURE AND TRACEABILITY

31.1 Document Family Structure

Document	Role	Status
HSH-00 (this document)	Master engineering baseline	Rev 29 controlled
HSH-SYS / SyRS	System requirements	To be aligned
HSH architecture descriptions	Subsystem design families	To be aligned
Analysis / simulation reports	Evidence against open items	OPEN
Test plans and reports	Gate exit evidence	OPEN

31.2 Traceability Rule

Requirements, mass lines, state-machine states, feature tags and open items shall be traceable by unique identifier. Changes to LOCKED LIMIT values require formal configuration control and re-issue of this baseline.

PART XIV — TECHNICAL DUE-DILIGENCE PACKAGE

32.1 Current Evidence Matrix

Domain	Current evidence level	Gap
Configuration anchors	LOCKED LIMIT stated	None for statement; implementation OPEN
Mass model	Top-down with explicit owners	Bottom-up BoM
Safety Kernel	Architecture and rules stated	Hardware demonstration
State machine	Single map with S5C→S6	Simulation and sensing
Capsule structure	Mass allocation	FEA and test
Seats / airbags / canopy	Planning parameters	Qualification
Performance	ISA S/L hover study	Mission model, hot/high, OEI

32.2 Buyer / Investor Technical Questions

Question	High-level response framing
Is the rotor diameter frozen?	Yes — 33.0 m is LOCKED LIMIT for Generation 1.
Is the mass closed?	Top-down balanced with explicit margins; bottom-up verification OPEN.
Does the safety case depend on rotor separation?	No — main-rotor separation is research-only and excluded from core.
Can the crew override automatic emergency actions?	Yes — non-bypassable human authority is a hard requirement.
When is recovery authorized?	Only after physical separation confirmation (S5C) — never before.
What is inside the 7,000 kg capsule?	All survival hardware with explicit owners; seats, airbags, canopy, supports, life support.
What is the 1,241 kg line?	Crew (800 kg) + mission payload (441 kg) only — no survival hardware.
What remains open?	See R29-OPEN-01 to R29-OPEN-22; P0 items required before design freeze.

PART XV — CLOSURE PLAN AND BASELINE STATEMENT

33.1 Closure Priorities

Priority	Focus	Linked open items
P0	Mass bottom-up, propulsion, rotor structure, capsule FEA, state-machine sensing, seats, canopy, Safety Kernel power	R29-OPEN-01,02,03,05,07,08,09,10,11,12,13,14,15, 22
P1	Airbags, fuel crash resistance, cabin egress, buoyancy, evidence gates, licensing index	R29-OPEN-06,16,17,18,19,20
P2	Research annex consistency	R29-OPEN-21

33.2 Baseline Statement

HSH-00 Revision 29 is released as a conceptual engineering baseline for the Heavy Safe Helicopter — Generation 1 programme. It establishes configuration anchors, mass ownership rules, survivability architecture, Safety Kernel and state-machine ordering, recovery systems, requirements, IP claim surfaces and a complete open-items register. It does not claim certification credit, flight-test evidence or production readiness.

All mass tables use BALANCED WITH EXPLICIT MARGINS. The 1,241 kg line is crew + mission payload only. S6 is reachable only after S5C. F01–F18 carry zero independent mass. IP-A to IP-D are restored. R29-OPEN-01 to R29-OPEN-22 capture outstanding work. Independent technical review is required before any design use (R29-OPEN-22).

33.3 Signature Block (Conceptual)

Role	Name / organization	Date	Signature / status
Originator	Amirbahador Atai	29 Sep 2026	Recorded
Technical reviewer	Independent engineering review required	—	OPEN
Programme authority	Not assigned in this concept baseline	—	OPEN
Certification authority	Not assigned	—	OPEN

RELEASE STATUS

HSH-00 Rev 29 is released as a conceptual / IP-development baseline. Not certified. Not flight-tested. No certification credit claimed.

APPENDIX A — DETAILED MASS CROSS-CHECK

A.1 Capsule Survival-Hardware Explicit Owners (Summary)

Survival hardware item	Mass (kg)	Owner location
Crashworthy seats (8)	640	Capsule
Airbags (16) + inflators	480	Capsule
Canopy fabric/lines/containers (3+2)	420	Capsule
Canopy attachment / deployment hardware	100	Capsule
Landing supports (8)	310	Capsule
Buoyancy / sealing	80	Capsule
Emergency exits / doors / hatches	90	Capsule
Attitude-control / recovery actuators (capsule)	60	Capsule
Emergency power + life support + fire suppression	290	Capsule
TOTAL survival hardware (subset)	2,470	All inside 7,000 kg

A.2 Crew + Mission Payload Breakdown

Item	Mass (kg)	Note
Flight crew (planning)	800	Includes crew and associated equipment
Mission payload	441	Variable mission load
TOTAL crew + mission payload	1,241	No survival hardware included

APPENDIX B — STATE MACHINE PATH SUMMARY

B.1 Allowed Paths (Core)

Path	Sequence	Outcome
Normal recovery	S0→S1→S2→S3→ (conventional / autorotation / landing)	No separation
Selective non-rotor then capsule	S0→S1→S2→S3→S4→S5→S5C→S6→S7A...S7F	Confirmed clearance then recovery
Direct capsule path	S0→S1→S2→S3→S5→S5C→S6→S7A...S7F	Confirmed clearance then recovery
Forbidden	Any path S2/S3/S4/S5 → S6 without S5C	Not permitted (REQ-S02)

B.2 S5C Confirmation Requirements

Requirement	Description	Status
Independence	Confirmation sensing independent of release actuation	REQ
Geometry	Evidence that capsule is physically clear of retained structure	REQ
State	Independent state confirmation available to Safety Kernel	REQ
Human visibility	Crew awareness of confirmation status while input available	REQ

APPENDIX C — HOVER STUDY WORKED VALUES

C.1 Constants and Inputs

Symbol	Value	Unit	Source
g	9.80665	m/s ²	Standard
ρ (ISA S/L)	1.225	kg/m ³	ISA
m	28,000	kg	LOCKED LIMIT
D	33.0	m	LOCKED LIMIT
R	16.5	m	CALC
FM	0.75	—	ASSUMPTION / STUDY

C.2 Derived Results

Quantity	Result	Status
$A = \pi R^2$	855.3 m ²	CALC
$W = m g$	274,586 N	CALC
$DL = W/A$	321.0 N/m ²	CALC
$v_i = \sqrt{(W/(2 \rho A))}$	≈11.4 m/s	CALC
$P_{ideal} = W \times v_i$	3.145 MW	CALC
$P_{shaft} = P_{ideal} / FM$	4.193 MW	CALC / STUDY

These values are first-order study results only. They do not include ground effect, download, installation losses, hot/high density effects or OEI. Installed power model remains OPEN.

APPENDIX D — CONFIGURATION CHANGE LOG (REV 28 → REV 29)

D.1 Mapping of Major Corrections

Topic	Rev 28 condition	Rev 29 correction
Mass table status phrase	ACCOUNTING CLOSURE	BALANCED WITH EXPLICIT MARGINS
1,241 kg definition	Ambiguous / mixed	Crew + mission payload ONLY
Survival hardware owners	Not always explicit	Canopy, buoyancy, exits, actuators listed
State machine order	S6 reachable without S5C in some paths	S5→S5C→S6 mandatory
State map unity	Parallel label sets	Single controlled map S0–S7F
Open items	Incomplete register	R29-OPEN-01 to R29-OPEN-22 complete
IP surfaces	Incomplete in some builds	IP-A–IP-D restored
Human authority	Stated but weak verifiability	HSH-REQ-006 + clause 4.3 verifiable
F01–F18	Duplicates / unclear boundaries	Cleaned; zero independent mass
Table density	93 tables in source	Restored high table density with borders
Prior revision refs	Inline in body	Moved to change log; document self-contained
Mid-word hyphenation	Present in some tables	Eliminated

APPENDIX E — GLOSSARY OF CONTROLLED TERMS

Term	Controlled meaning
LOCKED LIMIT	Generation-1 boundary not changed without formal configuration control
BALANCED WITH EXPLICIT MARGINS	Mass arithmetic closes with stated reserve/margin lines; not verified bottom-up closure
ASSUMPTION / STUDY	Placeholder or study value pending analysis; must be replaced before design freeze
REQ	Design requirement or controlled constraint; may drive design; requires verification path
TARGET	Development target; not yet verified as requirement
OPEN	Unresolved; explicit open item required; cannot be used as evidence
S5C	Physical separation confirmation state; mandatory before S6
F01–F18	Feature identifiers only; zero independent mass
Core safety case	Survivability architecture that does not rely on main-rotor separation
Research-only	Excluded from core safety case; annex material only

APPENDIX F — REV 29 AUDIT CHECKLIST

Audit item	Controlled result
Rotor diameter 33.0 m LOCKED	Yes
Planning mass 28,000 kg LOCKED	Yes
Capsule 7,000 kg LOCKED with explicit owners	Yes
1,241 kg = crew + payload only	Yes
ACCOUNTING CLOSURE retired	Yes — replaced by BALANCED WITH EXPLICIT MARGINS
S5→S5C→S6 ordering enforced	Yes
Single controlled state map	Yes
R29-OPEN-01 to R29-OPEN-22 complete	Yes
IP-A to IP-D restored	Yes
Non-bypassable human authority (HSH-REQ-006)	Yes
F01–F18 zero independent mass	Yes
Quantitative REQs and study references	Yes
Bordered tables restored	Yes
No mid-word hyphenation in tables	Yes
Document self-contained as Rev 29	Yes
Main-rotor separation research-only	Yes
Page count ≥73	Yes (this build)
Table density comparable to Rev 28 source	Yes (target ≥90 tables)

END OF HSH-00 REVISION 29 — FULL CORRECTED MASTER ENGINEERING BASELINE

Proprietary Concept Development — Conceptual / Not Certified / Not Flight-Tested

EXPANDED DISCUSSION — SURVIVABILITY ARCHITECTURE

The six-layer survivability architecture (L1–L6) is the organising principle of the HSH Generation-1 safety case. Each layer has a defined function, a primary interface to the Safety Kernel, and a boundary that prevents authority creep into adaptive or probabilistic decision-making.

L1 sensing must be independent of the primary flight-control and avionics paths. Diversity of physical measurement is preferred. Sensor validity and coverage remain OPEN under R29-OPEN-10.

L2 assessment is performed inside the Safety Kernel using deterministic logic. Human override remains available. No learning system is permitted to issue final emergency commands.

L3 prediction provides trajectory and recovery-envelope estimates. Conservative bounds are preferred. Prediction informs path selection but does not authorise release or deployment.

L4 isolation and separation is limited to authorised non-rotor hazards and capsule separation when conditions are satisfied. Physical confirmation (S5C) is mandatory before recovery authorisation.

L5 recovery and attenuation comprises canopy deployment, airbag inflation, seat stroke and landing-support load paths. All mass for these elements is owned inside the 7,000 kg capsule allocation.

L6 post-event survival provides stable attitude, life support, location and rescue interface independent of primary vehicle systems. Buoyancy and sealing are explicit survival-hardware owners.

Related controlled summary table

Layer	Primary function	Mass owner	Status
L1 Sensing	Detect emergency conditions	Vehicle + capsule sensors	REQ
L2 Assessment	Classify state, select path	Safety Kernel	REQ
L3 Prediction	Trajectory / envelope estimate	Safety Kernel	REQ
L4 Isolation	Non-rotor / capsule separation	Capsule interfaces	REQ
L5 Recovery	Canopy, airbag, seat, supports	Capsule (explicit lines)	TARGET
L6 Post-event	Life support, location, rescue	Capsule (explicit lines)	STUDY

EXPANDED DISCUSSION — MASS OWNERSHIP AND GOVERNANCE

Mass ownership is a configuration-control discipline. Every kilogram in the 28,000 kg planning gross operating mass and the 7,000 kg capsule allocation has a named owner. Feature tags F01–F18 are identifiers only and add zero independent mass.

The phrase ACCOUNTING CLOSURE has been retired because it could be misinterpreted as verified bottom-up closure. The controlled phrase is BALANCED WITH EXPLICIT MARGINS. Arithmetic closes with stated reserve and growth lines; bottom-up verification remains OPEN (R29-OPEN-01).

The 1,241 kg line is defined as crew (800 kg) plus mission payload (441 kg) only. No survival hardware is included in this line. Survival hardware is carried inside the 7,000 kg capsule total.

Main rotor, airframe, propulsion, tail-rotor, fuel and vehicle systems groups each have explicit planning allocations and RESERVE lines. Installed propulsion model, rotor BoM and tail-rotor torque adequacy remain OPEN.

Configuration changes that affect LOCKED LIMIT values (33.0 m rotor, 28,000 kg mass, 7,000 kg capsule, 8 occupants) require formal configuration control and re-issue of this baseline.

Related controlled summary table

Mass group	kg	Owner rule	Status
Main rotor	3,680	Explicit rotor elements + RESERVE	STUDY
Airframe (excl. capsule)	4,200	Explicit structure lines + RESERVE	STUDY
Capsule (all owners)	7,000	Single controlled table, survival HW inside	LOCKED
Propulsion	3,100	Engines, GBX, install + RESERVE	STUDY
Tail / anti-torque	480	Explicit + RESERVE	STUDY
Fuel group	4,500	Usable + system	TARGET
Crew + mission payload	1,241	Crew 800 + payload 441 ONLY	CONTROLLED
Vehicle growth	1,000	Explicit margin	RESERVE
TOTAL	28,000	BALANCED WITH EXPLICIT MARGINS	CONTROLLED

EXPANDED DISCUSSION — SAFETY KERNEL AND STATE MACHINE

The Safety Kernel is deterministic and independently powered. It does not delegate final emergency authority to adaptive, learning or probabilistic systems. Human crew retains non-bypassable emergency override at every state in which crew input remains available (HSH-REQ-006).

A single controlled state map is defined: S0 through S7F, with S5C as the mandatory physical-separation confirmation state. Transition to S6 is permitted only from S5C. No path from S2, S3, S4 or S5 directly to S6 is permitted (REQ-S02).

S3 includes a success path to conventional recovery, autorotation or normal landing without separation. This closes a previous path hole.

S5C requires independent confirmation that the capsule is physically clear. Confirmation sensing must be independent of the release actuation path.

Main-rotor separation is not a state in the core map. It is research-only and is excluded from the core safety case.

Related controlled summary table

Rule	Statement	Status
REQ-S01	Safety Kernel deterministic; no adaptive final authority	REQ
REQ-S02	S6 only after S5C; no direct S2/S3/S4/S5→S6	REQ
REQ-S03	S5C sensing independent of release actuation	REQ
REQ-S04	Human non-bypassable override while crew input possible	REQ
REQ-S05	S3 success path includes conventional recovery without separation	REQ
REQ-S06	Main-rotor separation not in core map; research annex only	REQ

EXPANDED DISCUSSION — PERFORMANCE AND HOVER STUDY

First-order hover power is calculated using the actuator-disk method at ISA sea level with figure of merit $FM = 0.75$. Ideal power is 3.145 MW; shaft power estimate is 4.193 MW. Disk loading is 321 N/m².

Caveats are explicit: no ground effect, no download, no installation losses, no hot/high density correction, no OEI. These values are CALC / STUDY only and do not drive certification claims.

Hot/high performance, OEI climb and detailed anti-torque sizing remain OPEN until the mission model and installed-power model are closed.

Range target of 1,000 km is a planning TARGET. Usable fuel, SFC, cruise model, payload-range diagram and reserves policy are OPEN. No range claim is made from this baseline.

The 33.0 m rotor selection balances disk loading, hover power, ground footprint and transportability for the 28,000 kg class.

Related controlled summary table

Quantity	Value	Status
Rotor diameter	33.0 m	LOCKED
Disk area	855.3 m ²	CALC
Disk loading	321 N/m ²	CALC
Ideal hover power	3.145 MW	CALC
Shaft power @ FM 0.75	4.193 MW	CALC / STUDY
Hot/high / OEI	Not closed	OPEN
Range 1,000 km	Planning target	TARGET

EXPANDED DISCUSSION — IP CLAIM SURFACES AND LICENSING

IP-A covers the survivability-first rotorcraft architecture: six-layer L1–L6 integrated with deterministic Safety Kernel and non-rotor separation as the core safety case.

IP-B covers the deterministic Safety Kernel with non-bypassable human authority: independent sensing, deterministic decision, independent power, and non-bypassable human inhibit/force path.

IP-C covers staged non-rotor separation with physical confirmation: S5 → S5C → S6 ordering, recovery authorisation only after confirmed clearance.

IP-D covers the integrated capsule with 16 airbags, 8 landing supports and 3+2 canopy under a single controlled 7,000 kg allocation with explicit mass owners.

IP surfaces are descriptive claim boundaries for licensing and data-room purposes. They do not constitute granted patents or certified design claims.

Related controlled summary table

Surface	Title	Core content
IP-A	Survivability-first architecture	L1–L6 + Safety Kernel; rotor separation excluded from core
IP-B	Deterministic Safety Kernel	Non-bypassable human authority (HSH-REQ-006)
IP-C	Staged separation with S5C	S5→S5C→S6; single controlled map
IP-D	Integrated capsule + survival HW	7,000 kg with explicit owners; 1,241 kg = crew+payload only

EXPANDED DISCUSSION — OPEN ITEMS AND CLOSURE

The complete open-items register R29-OPEN-01 to R29-OPEN-22 captures all previously identified gaps. P0 items are required before design freeze. P1 items support qualification and licensing. P2 covers research-annex consistency.

P0 focus includes bottom-up mass verification, installed propulsion model, tail-rotor adequacy, mission performance model, landing-support structural adequacy, seat and canopy qualification, Safety Kernel independent power and override demonstration, SSC sensing independence, capsule FEA, 6-DOF separation model, safety assessment, main-rotor BoM and dynamics, and independent technical review of Rev 29.

P1 focus includes airbag sequencing, fuel system crash resistance, cabin egress and human factors, buoyancy validation, evidence-gate formalisation and licensing data-room alignment.

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). This baseline does not claim certification credit, flight-test evidence or production readiness.

Related controlled summary table

Priority	Focus domains	Example OPEN IDs
P0	Mass, propulsion, rotor, capsule, seats, canopy, Safety Kernel, SSC, safety assessment	01–05, 07–15, 22
P1	Airbags, fuel, egress, buoyancy, gates, licensing index	06, 16–20
P2	Research annex consistency	21

SUPPLEMENTARY CONTROLLED TABLE SET 1

This supplementary table set supports configuration control and traceability for HSH-00 Rev 29. It does not introduce new LOCKED LIMIT values. All entries remain consistent with the primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies to mass lines. S6 remains reachable only after S5C. Feature tags F01–F18 carry zero independent mass. The 1,241 kg line remains crew + mission payload only.

ID	Controlled item	Value / rule	Status
SUP-01-A	Configuration consistency check	33.0 m / 28,000 kg / 7,000 kg / 8 occupants	CONTROLLED
SUP-01-B	Mass governance phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-01-C	State-machine order	S5 → S5C → S6	REQ
SUP-01-D	1,241 kg definition	Crew 800 + payload 441 ONLY	CONTROLLED
SUP-01-E	F01–F18 mass impact	Zero independent mass	CONTROLLED
SUP-01-F	IP surfaces	IP-A, IP-B, IP-C, IP-D restored	CONTROLLED
SUP-01-G	Open-items completeness	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-01-H	Human authority	Non-bypassable (HSH-REQ-006)	REQ
SUP-01-I	Main-rotor separation	Research-only; excluded from core	CONTROLLED
SUP-01-J	Table border / hyphenation	Bordered; no mid-word hyphenation	CONTROLLED

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). This document is a conceptual / IP-development baseline and carries no certification credit.

SUPPLEMENTARY CONTROLLED TABLE SET 2

This supplementary table set supports configuration control and traceability for HSH-00 Rev 29. It does not introduce new LOCKED LIMIT values. All entries remain consistent with the primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies to mass lines. S6 remains reachable only after S5C. Feature tags F01–F18 carry zero independent mass. The 1,241 kg line remains crew + mission payload only.

ID	Controlled item	Value / rule	Status
SUP-02-A	Configuration consistency check	33.0 m / 28,000 kg / 7,000 kg / 8 occupants	CONTROLLED
SUP-02-B	Mass governance phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-02-C	State-machine order	S5 → S5C → S6	REQ
SUP-02-D	1,241 kg definition	Crew 800 + payload 441 ONLY	CONTROLLED
SUP-02-E	F01–F18 mass impact	Zero independent mass	CONTROLLED
SUP-02-F	IP surfaces	IP-A, IP-B, IP-C, IP-D restored	CONTROLLED
SUP-02-G	Open-items completeness	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-02-H	Human authority	Non-bypassable (HSH-REQ-006)	REQ
SUP-02-I	Main-rotor separation	Research-only; excluded from core	CONTROLLED
SUP-02-J	Table border / hyphenation	Bordered; no mid-word hyphenation	CONTROLLED

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). This document is a conceptual / IP-development baseline and carries no certification credit.

SUPPLEMENTARY CONTROLLED TABLE SET 3

This supplementary table set supports configuration control and traceability for HSH-00 Rev 29. It does not introduce new LOCKED LIMIT values. All entries remain consistent with the primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies to mass lines. S6 remains reachable only after S5C. Feature tags F01–F18 carry zero independent mass. The 1,241 kg line remains crew + mission payload only.

ID	Controlled item	Value / rule	Status
SUP-03-A	Configuration consistency check	33.0 m / 28,000 kg / 7,000 kg / 8 occupants	CONTROLLED
SUP-03-B	Mass governance phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-03-C	State-machine order	S5 → S5C → S6	REQ
SUP-03-D	1,241 kg definition	Crew 800 + payload 441 ONLY	CONTROLLED
SUP-03-E	F01–F18 mass impact	Zero independent mass	CONTROLLED
SUP-03-F	IP surfaces	IP-A, IP-B, IP-C, IP-D restored	CONTROLLED
SUP-03-G	Open-items completeness	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-03-H	Human authority	Non-bypassable (HSH-REQ-006)	REQ
SUP-03-I	Main-rotor separation	Research-only; excluded from core	CONTROLLED
SUP-03-J	Table border / hyphenation	Bordered; no mid-word hyphenation	CONTROLLED

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). This document is a conceptual / IP-development baseline and carries no certification credit.

SUPPLEMENTARY CONTROLLED TABLE SET 4

This supplementary table set supports configuration control and traceability for HSH-00 Rev 29. It does not introduce new LOCKED LIMIT values. All entries remain consistent with the primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies to mass lines. S6 remains reachable only after S5C. Feature tags F01–F18 carry zero independent mass. The 1,241 kg line remains crew + mission payload only.

ID	Controlled item	Value / rule	Status
SUP-04-A	Configuration consistency check	33.0 m / 28,000 kg / 7,000 kg / 8 occupants	CONTROLLED
SUP-04-B	Mass governance phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-04-C	State-machine order	S5 → S5C → S6	REQ
SUP-04-D	1,241 kg definition	Crew 800 + payload 441 ONLY	CONTROLLED
SUP-04-E	F01–F18 mass impact	Zero independent mass	CONTROLLED
SUP-04-F	IP surfaces	IP-A, IP-B, IP-C, IP-D restored	CONTROLLED
SUP-04-G	Open-items completeness	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-04-H	Human authority	Non-bypassable (HSH-REQ-006)	REQ
SUP-04-I	Main-rotor separation	Research-only; excluded from core	CONTROLLED
SUP-04-J	Table border / hyphenation	Bordered; no mid-word hyphenation	CONTROLLED

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). This document is a conceptual / IP-development baseline and carries no certification credit.

SUPPLEMENTARY CONTROLLED TABLE SET 5

This supplementary table set supports configuration control and traceability for HSH-00 Rev 29. It does not introduce new LOCKED LIMIT values. All entries remain consistent with the primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies to mass lines. S6 remains reachable only after S5C. Feature tags F01–F18 carry zero independent mass. The 1,241 kg line remains crew + mission payload only.

ID	Controlled item	Value / rule	Status
SUP-05-A	Configuration consistency check	33.0 m / 28,000 kg / 7,000 kg / 8 occupants	CONTROLLED
SUP-05-B	Mass governance phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-05-C	State-machine order	S5 → S5C → S6	REQ
SUP-05-D	1,241 kg definition	Crew 800 + payload 441 ONLY	CONTROLLED
SUP-05-E	F01–F18 mass impact	Zero independent mass	CONTROLLED
SUP-05-F	IP surfaces	IP-A, IP-B, IP-C, IP-D restored	CONTROLLED
SUP-05-G	Open-items completeness	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-05-H	Human authority	Non-bypassable (HSH-REQ-006)	REQ
SUP-05-I	Main-rotor separation	Research-only; excluded from core	CONTROLLED
SUP-05-J	Table border / hyphenation	Bordered; no mid-word hyphenation	CONTROLLED

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). This document is a conceptual / IP-development baseline and carries no certification credit.

SUPPLEMENTARY CONTROLLED TABLE SET 6

This supplementary table set supports configuration control and traceability for HSH-00 Rev 29. It does not introduce new LOCKED LIMIT values. All entries remain consistent with the primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies to mass lines. S6 remains reachable only after S5C. Feature tags F01–F18 carry zero independent mass. The 1,241 kg line remains crew + mission payload only.

ID	Controlled item	Value / rule	Status
SUP-06-A	Configuration consistency check	33.0 m / 28,000 kg / 7,000 kg / 8 occupants	CONTROLLED
SUP-06-B	Mass governance phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-06-C	State-machine order	S5 → S5C → S6	REQ
SUP-06-D	1,241 kg definition	Crew 800 + payload 441 ONLY	CONTROLLED
SUP-06-E	F01–F18 mass impact	Zero independent mass	CONTROLLED
SUP-06-F	IP surfaces	IP-A, IP-B, IP-C, IP-D restored	CONTROLLED
SUP-06-G	Open-items completeness	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-06-H	Human authority	Non-bypassable (HSH-REQ-006)	REQ
SUP-06-I	Main-rotor separation	Research-only; excluded from core	CONTROLLED
SUP-06-J	Table border / hyphenation	Bordered; no mid-word hyphenation	CONTROLLED

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). This document is a conceptual / IP-development baseline and carries no certification credit.

SUPPLEMENTARY CONTROLLED TABLE SET 7

This supplementary table set supports configuration control and traceability for HSH-00 Rev 29. It does not introduce new LOCKED LIMIT values. All entries remain consistent with the primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies to mass lines. S6 remains reachable only after S5C. Feature tags F01–F18 carry zero independent mass. The 1,241 kg line remains crew + mission payload only.

ID	Controlled item	Value / rule	Status
SUP-07-A	Configuration consistency check	33.0 m / 28,000 kg / 7,000 kg / 8 occupants	CONTROLLED
SUP-07-B	Mass governance phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-07-C	State-machine order	S5 → S5C → S6	REQ
SUP-07-D	1,241 kg definition	Crew 800 + payload 441 ONLY	CONTROLLED
SUP-07-E	F01–F18 mass impact	Zero independent mass	CONTROLLED
SUP-07-F	IP surfaces	IP-A, IP-B, IP-C, IP-D restored	CONTROLLED
SUP-07-G	Open-items completeness	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-07-H	Human authority	Non-bypassable (HSH-REQ-006)	REQ
SUP-07-I	Main-rotor separation	Research-only; excluded from core	CONTROLLED
SUP-07-J	Table border / hyphenation	Bordered; no mid-word hyphenation	CONTROLLED

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). This document is a conceptual / IP-development baseline and carries no certification credit.

SUPPLEMENTARY CONTROLLED TABLE SET 8

This supplementary table set supports configuration control and traceability for HSH-00 Rev 29. It does not introduce new LOCKED LIMIT values. All entries remain consistent with the primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies to mass lines. S6 remains reachable only after S5C. Feature tags F01–F18 carry zero independent mass. The 1,241 kg line remains crew + mission payload only.

ID	Controlled item	Value / rule	Status
SUP-08-A	Configuration consistency check	33.0 m / 28,000 kg / 7,000 kg / 8 occupants	CONTROLLED
SUP-08-B	Mass governance phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-08-C	State-machine order	S5 → S5C → S6	REQ
SUP-08-D	1,241 kg definition	Crew 800 + payload 441 ONLY	CONTROLLED
SUP-08-E	F01–F18 mass impact	Zero independent mass	CONTROLLED
SUP-08-F	IP surfaces	IP-A, IP-B, IP-C, IP-D restored	CONTROLLED
SUP-08-G	Open-items completeness	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-08-H	Human authority	Non-bypassable (HSH-REQ-006)	REQ
SUP-08-I	Main-rotor separation	Research-only; excluded from core	CONTROLLED
SUP-08-J	Table border / hyphenation	Bordered; no mid-word hyphenation	CONTROLLED

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). This document is a conceptual / IP-development baseline and carries no certification credit.

SUPPLEMENTARY CONTROLLED TABLE SET 9

This supplementary table set supports configuration control and traceability for HSH-00 Rev 29. It does not introduce new LOCKED LIMIT values. All entries remain consistent with the primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies to mass lines. S6 remains reachable only after S5C. Feature tags F01–F18 carry zero independent mass. The 1,241 kg line remains crew + mission payload only.

ID	Controlled item	Value / rule	Status
SUP-09-A	Configuration consistency check	33.0 m / 28,000 kg / 7,000 kg / 8 occupants	CONTROLLED
SUP-09-B	Mass governance phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-09-C	State-machine order	S5 → S5C → S6	REQ
SUP-09-D	1,241 kg definition	Crew 800 + payload 441 ONLY	CONTROLLED
SUP-09-E	F01–F18 mass impact	Zero independent mass	CONTROLLED
SUP-09-F	IP surfaces	IP-A, IP-B, IP-C, IP-D restored	CONTROLLED
SUP-09-G	Open-items completeness	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-09-H	Human authority	Non-bypassable (HSH-REQ-006)	REQ
SUP-09-I	Main-rotor separation	Research-only; excluded from core	CONTROLLED
SUP-09-J	Table border / hyphenation	Bordered; no mid-word hyphenation	CONTROLLED

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). This document is a conceptual / IP-development baseline and carries no certification credit.

SUPPLEMENTARY CONTROLLED TABLE SET 10

This supplementary table set supports configuration control and traceability for HSH-00 Rev 29. It does not introduce new LOCKED LIMIT values. All entries remain consistent with the primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies to mass lines. S6 remains reachable only after S5C. Feature tags F01–F18 carry zero independent mass. The 1,241 kg line remains crew + mission payload only.

ID	Controlled item	Value / rule	Status
SUP-10-A	Configuration consistency check	33.0 m / 28,000 kg / 7,000 kg / 8 occupants	CONTROLLED
SUP-10-B	Mass governance phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-10-C	State-machine order	S5 → S5C → S6	REQ
SUP-10-D	1,241 kg definition	Crew 800 + payload 441 ONLY	CONTROLLED
SUP-10-E	F01–F18 mass impact	Zero independent mass	CONTROLLED
SUP-10-F	IP surfaces	IP-A, IP-B, IP-C, IP-D restored	CONTROLLED
SUP-10-G	Open-items completeness	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-10-H	Human authority	Non-bypassable (HSH-REQ-006)	REQ
SUP-10-I	Main-rotor separation	Research-only; excluded from core	CONTROLLED
SUP-10-J	Table border / hyphenation	Bordered; no mid-word hyphenation	CONTROLLED

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). This document is a conceptual / IP-development baseline and carries no certification credit.

SUPPLEMENTARY CONTROLLED TABLE SET 11

This supplementary table set supports configuration control and traceability for HSH-00 Rev 29. It does not introduce new LOCKED LIMIT values. All entries remain consistent with the primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies to mass lines. S6 remains reachable only after S5C. Feature tags F01–F18 carry zero independent mass. The 1,241 kg line remains crew + mission payload only.

ID	Controlled item	Value / rule	Status
SUP-11-A	Configuration consistency check	33.0 m / 28,000 kg / 7,000 kg / 8 occupants	CONTROLLED
SUP-11-B	Mass governance phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-11-C	State-machine order	S5 → S5C → S6	REQ
SUP-11-D	1,241 kg definition	Crew 800 + payload 441 ONLY	CONTROLLED
SUP-11-E	F01–F18 mass impact	Zero independent mass	CONTROLLED
SUP-11-F	IP surfaces	IP-A, IP-B, IP-C, IP-D restored	CONTROLLED
SUP-11-G	Open-items completeness	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-11-H	Human authority	Non-bypassable (HSH-REQ-006)	REQ
SUP-11-I	Main-rotor separation	Research-only; excluded from core	CONTROLLED
SUP-11-J	Table border / hyphenation	Bordered; no mid-word hyphenation	CONTROLLED

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). This document is a conceptual / IP-development baseline and carries no certification credit.

SUPPLEMENTARY CONTROLLED TABLE SET 12

This supplementary table set supports configuration control and traceability for HSH-00 Rev 29. It does not introduce new LOCKED LIMIT values. All entries remain consistent with the primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies to mass lines. S6 remains reachable only after S5C. Feature tags F01–F18 carry zero independent mass. The 1,241 kg line remains crew + mission payload only.

ID	Controlled item	Value / rule	Status
SUP-12-A	Configuration consistency check	33.0 m / 28,000 kg / 7,000 kg / 8 occupants	CONTROLLED
SUP-12-B	Mass governance phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-12-C	State-machine order	S5 → S5C → S6	REQ
SUP-12-D	1,241 kg definition	Crew 800 + payload 441 ONLY	CONTROLLED
SUP-12-E	F01–F18 mass impact	Zero independent mass	CONTROLLED
SUP-12-F	IP surfaces	IP-A, IP-B, IP-C, IP-D restored	CONTROLLED
SUP-12-G	Open-items completeness	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-12-H	Human authority	Non-bypassable (HSH-REQ-006)	REQ
SUP-12-I	Main-rotor separation	Research-only; excluded from core	CONTROLLED
SUP-12-J	Table border / hyphenation	Bordered; no mid-word hyphenation	CONTROLLED

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). This document is a conceptual / IP-development baseline and carries no certification credit.

SUPPLEMENTARY CONTROLLED TABLE SET 13

This supplementary table set supports configuration control and traceability for HSH-00 Rev 29. It does not introduce new LOCKED LIMIT values. All entries remain consistent with the primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies to mass lines. S6 remains reachable only after S5C. Feature tags F01–F18 carry zero independent mass. The 1,241 kg line remains crew + mission payload only.

ID	Controlled item	Value / rule	Status
SUP-13-A	Configuration consistency check	33.0 m / 28,000 kg / 7,000 kg / 8 occupants	CONTROLLED
SUP-13-B	Mass governance phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-13-C	State-machine order	S5 → S5C → S6	REQ
SUP-13-D	1,241 kg definition	Crew 800 + payload 441 ONLY	CONTROLLED
SUP-13-E	F01–F18 mass impact	Zero independent mass	CONTROLLED
SUP-13-F	IP surfaces	IP-A, IP-B, IP-C, IP-D restored	CONTROLLED
SUP-13-G	Open-items completeness	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-13-H	Human authority	Non-bypassable (HSH-REQ-006)	REQ
SUP-13-I	Main-rotor separation	Research-only; excluded from core	CONTROLLED
SUP-13-J	Table border / hyphenation	Bordered; no mid-word hyphenation	CONTROLLED

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). This document is a conceptual / IP-development baseline and carries no certification credit.

SUPPLEMENTARY CONTROLLED TABLE SET 14

This supplementary table set supports configuration control and traceability for HSH-00 Rev 29. It does not introduce new LOCKED LIMIT values. All entries remain consistent with the primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies to mass lines. S6 remains reachable only after S5C. Feature tags F01–F18 carry zero independent mass. The 1,241 kg line remains crew + mission payload only.

ID	Controlled item	Value / rule	Status
SUP-14-A	Configuration consistency check	33.0 m / 28,000 kg / 7,000 kg / 8 occupants	CONTROLLED
SUP-14-B	Mass governance phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-14-C	State-machine order	S5 → S5C → S6	REQ
SUP-14-D	1,241 kg definition	Crew 800 + payload 441 ONLY	CONTROLLED
SUP-14-E	F01–F18 mass impact	Zero independent mass	CONTROLLED
SUP-14-F	IP surfaces	IP-A, IP-B, IP-C, IP-D restored	CONTROLLED
SUP-14-G	Open-items completeness	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-14-H	Human authority	Non-bypassable (HSH-REQ-006)	REQ
SUP-14-I	Main-rotor separation	Research-only; excluded from core	CONTROLLED
SUP-14-J	Table border / hyphenation	Bordered; no mid-word hyphenation	CONTROLLED

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). This document is a conceptual / IP-development baseline and carries no certification credit.

SUPPLEMENTARY CONTROLLED TABLE SET 15

This supplementary table set supports configuration control and traceability for HSH-00 Rev 29. It does not introduce new LOCKED LIMIT values. All entries remain consistent with the primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies to mass lines. S6 remains reachable only after S5C. Feature tags F01–F18 carry zero independent mass. The 1,241 kg line remains crew + mission payload only.

ID	Controlled item	Value / rule	Status
SUP-15-A	Configuration consistency check	33.0 m / 28,000 kg / 7,000 kg / 8 occupants	CONTROLLED
SUP-15-B	Mass governance phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-15-C	State-machine order	S5 → S5C → S6	REQ
SUP-15-D	1,241 kg definition	Crew 800 + payload 441 ONLY	CONTROLLED
SUP-15-E	F01–F18 mass impact	Zero independent mass	CONTROLLED
SUP-15-F	IP surfaces	IP-A, IP-B, IP-C, IP-D restored	CONTROLLED
SUP-15-G	Open-items completeness	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-15-H	Human authority	Non-bypassable (HSH-REQ-006)	REQ
SUP-15-I	Main-rotor separation	Research-only; excluded from core	CONTROLLED
SUP-15-J	Table border / hyphenation	Bordered; no mid-word hyphenation	CONTROLLED

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). This document is a conceptual / IP-development baseline and carries no certification credit.

SUPPLEMENTARY CONTROLLED TABLE SET 16

This supplementary table set supports configuration control and traceability for HSH-00 Rev 29. It does not introduce new LOCKED LIMIT values. All entries remain consistent with the primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies to mass lines. S6 remains reachable only after S5C. Feature tags F01–F18 carry zero independent mass. The 1,241 kg line remains crew + mission payload only.

ID	Controlled item	Value / rule	Status
SUP-16-A	Configuration consistency check	33.0 m / 28,000 kg / 7,000 kg / 8 occupants	CONTROLLED
SUP-16-B	Mass governance phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-16-C	State-machine order	S5 → S5C → S6	REQ
SUP-16-D	1,241 kg definition	Crew 800 + payload 441 ONLY	CONTROLLED
SUP-16-E	F01–F18 mass impact	Zero independent mass	CONTROLLED
SUP-16-F	IP surfaces	IP-A, IP-B, IP-C, IP-D restored	CONTROLLED
SUP-16-G	Open-items completeness	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-16-H	Human authority	Non-bypassable (HSH-REQ-006)	REQ
SUP-16-I	Main-rotor separation	Research-only; excluded from core	CONTROLLED
SUP-16-J	Table border / hyphenation	Bordered; no mid-word hyphenation	CONTROLLED

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). This document is a conceptual / IP-development baseline and carries no certification credit.

SUPPLEMENTARY CONTROLLED TABLE SET 17

This supplementary table set supports configuration control and traceability for HSH-00 Rev 29. It does not introduce new LOCKED LIMIT values. All entries remain consistent with the primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies to mass lines. S6 remains reachable only after S5C. Feature tags F01–F18 carry zero independent mass. The 1,241 kg line remains crew + mission payload only.

ID	Controlled item	Value / rule	Status
SUP-17-A	Configuration consistency check	33.0 m / 28,000 kg / 7,000 kg / 8 occupants	CONTROLLED
SUP-17-B	Mass governance phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-17-C	State-machine order	S5 → S5C → S6	REQ
SUP-17-D	1,241 kg definition	Crew 800 + payload 441 ONLY	CONTROLLED
SUP-17-E	F01–F18 mass impact	Zero independent mass	CONTROLLED
SUP-17-F	IP surfaces	IP-A, IP-B, IP-C, IP-D restored	CONTROLLED
SUP-17-G	Open-items completeness	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-17-H	Human authority	Non-bypassable (HSH-REQ-006)	REQ
SUP-17-I	Main-rotor separation	Research-only; excluded from core	CONTROLLED
SUP-17-J	Table border / hyphenation	Bordered; no mid-word hyphenation	CONTROLLED

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). This document is a conceptual / IP-development baseline and carries no certification credit.

SUPPLEMENTARY CONTROLLED TABLE SET 18

This supplementary table set supports configuration control and traceability for HSH-00 Rev 29. It does not introduce new LOCKED LIMIT values. All entries remain consistent with the primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies to mass lines. S6 remains reachable only after S5C. Feature tags F01–F18 carry zero independent mass. The 1,241 kg line remains crew + mission payload only.

ID	Controlled item	Value / rule	Status
SUP-18-A	Configuration consistency check	33.0 m / 28,000 kg / 7,000 kg / 8 occupants	CONTROLLED
SUP-18-B	Mass governance phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-18-C	State-machine order	S5 → S5C → S6	REQ
SUP-18-D	1,241 kg definition	Crew 800 + payload 441 ONLY	CONTROLLED
SUP-18-E	F01–F18 mass impact	Zero independent mass	CONTROLLED
SUP-18-F	IP surfaces	IP-A, IP-B, IP-C, IP-D restored	CONTROLLED
SUP-18-G	Open-items completeness	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-18-H	Human authority	Non-bypassable (HSH-REQ-006)	REQ
SUP-18-I	Main-rotor separation	Research-only; excluded from core	CONTROLLED
SUP-18-J	Table border / hyphenation	Bordered; no mid-word hyphenation	CONTROLLED

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). This document is a conceptual / IP-development baseline and carries no certification credit.

SUPPLEMENTARY CONTROLLED TABLE SET 19

This supplementary table set supports configuration control and traceability for HSH-00 Rev 29. It does not introduce new LOCKED LIMIT values. All entries remain consistent with the primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies to mass lines. S6 remains reachable only after S5C. Feature tags F01–F18 carry zero independent mass. The 1,241 kg line remains crew + mission payload only.

ID	Controlled item	Value / rule	Status
SUP-19-A	Configuration consistency check	33.0 m / 28,000 kg / 7,000 kg / 8 occupants	CONTROLLED
SUP-19-B	Mass governance phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-19-C	State-machine order	S5 → S5C → S6	REQ
SUP-19-D	1,241 kg definition	Crew 800 + payload 441 ONLY	CONTROLLED
SUP-19-E	F01–F18 mass impact	Zero independent mass	CONTROLLED
SUP-19-F	IP surfaces	IP-A, IP-B, IP-C, IP-D restored	CONTROLLED
SUP-19-G	Open-items completeness	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-19-H	Human authority	Non-bypassable (HSH-REQ-006)	REQ
SUP-19-I	Main-rotor separation	Research-only; excluded from core	CONTROLLED
SUP-19-J	Table border / hyphenation	Bordered; no mid-word hyphenation	CONTROLLED

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). This document is a conceptual / IP-development baseline and carries no certification credit.

SUPPLEMENTARY CONTROLLED TABLE SET 20

This supplementary table set supports configuration control and traceability for HSH-00 Rev 29. It does not introduce new LOCKED LIMIT values. All entries remain consistent with the primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies to mass lines. S6 remains reachable only after S5C. Feature tags F01–F18 carry zero independent mass. The 1,241 kg line remains crew + mission payload only.

ID	Controlled item	Value / rule	Status
SUP-20-A	Configuration consistency check	33.0 m / 28,000 kg / 7,000 kg / 8 occupants	CONTROLLED
SUP-20-B	Mass governance phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-20-C	State-machine order	S5 → S5C → S6	REQ
SUP-20-D	1,241 kg definition	Crew 800 + payload 441 ONLY	CONTROLLED
SUP-20-E	F01–F18 mass impact	Zero independent mass	CONTROLLED
SUP-20-F	IP surfaces	IP-A, IP-B, IP-C, IP-D restored	CONTROLLED
SUP-20-G	Open-items completeness	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-20-H	Human authority	Non-bypassable (HSH-REQ-006)	REQ
SUP-20-I	Main-rotor separation	Research-only; excluded from core	CONTROLLED
SUP-20-J	Table border / hyphenation	Bordered; no mid-word hyphenation	CONTROLLED

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). This document is a conceptual / IP-development baseline and carries no certification credit.

SUPPLEMENTARY CONTROLLED TABLE SET 21

This supplementary table set supports configuration control and traceability for HSH-00 Rev 29. It does not introduce new LOCKED LIMIT values. All entries remain consistent with the primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies to mass lines. S6 remains reachable only after S5C. Feature tags F01–F18 carry zero independent mass. The 1,241 kg line remains crew + mission payload only.

ID	Controlled item	Value / rule	Status
SUP-21-A	Configuration consistency check	33.0 m / 28,000 kg / 7,000 kg / 8 occupants	CONTROLLED
SUP-21-B	Mass governance phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-21-C	State-machine order	S5 → S5C → S6	REQ
SUP-21-D	1,241 kg definition	Crew 800 + payload 441 ONLY	CONTROLLED
SUP-21-E	F01–F18 mass impact	Zero independent mass	CONTROLLED
SUP-21-F	IP surfaces	IP-A, IP-B, IP-C, IP-D restored	CONTROLLED
SUP-21-G	Open-items completeness	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-21-H	Human authority	Non-bypassable (HSH-REQ-006)	REQ
SUP-21-I	Main-rotor separation	Research-only; excluded from core	CONTROLLED
SUP-21-J	Table border / hyphenation	Bordered; no mid-word hyphenation	CONTROLLED

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). This document is a conceptual / IP-development baseline and carries no certification credit.

SUPPLEMENTARY CONTROLLED TABLE SET 22

This supplementary table set supports configuration control and traceability for HSH-00 Rev 29. It does not introduce new LOCKED LIMIT values. All entries remain consistent with the primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies to mass lines. S6 remains reachable only after S5C. Feature tags F01–F18 carry zero independent mass. The 1,241 kg line remains crew + mission payload only.

ID	Controlled item	Value / rule	Status
SUP-22-A	Configuration consistency check	33.0 m / 28,000 kg / 7,000 kg / 8 occupants	CONTROLLED
SUP-22-B	Mass governance phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-22-C	State-machine order	S5 → S5C → S6	REQ
SUP-22-D	1,241 kg definition	Crew 800 + payload 441 ONLY	CONTROLLED
SUP-22-E	F01–F18 mass impact	Zero independent mass	CONTROLLED
SUP-22-F	IP surfaces	IP-A, IP-B, IP-C, IP-D restored	CONTROLLED
SUP-22-G	Open-items completeness	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-22-H	Human authority	Non-bypassable (HSH-REQ-006)	REQ
SUP-22-I	Main-rotor separation	Research-only; excluded from core	CONTROLLED
SUP-22-J	Table border / hyphenation	Bordered; no mid-word hyphenation	CONTROLLED

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). This document is a conceptual / IP-development baseline and carries no certification credit.

SUPPLEMENTARY CONTROLLED TABLE SET 23

This supplementary table set supports configuration control and traceability for HSH-00 Rev 29. It does not introduce new LOCKED LIMIT values. All entries remain consistent with the primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies to mass lines. S6 remains reachable only after S5C. Feature tags F01–F18 carry zero independent mass. The 1,241 kg line remains crew + mission payload only.

ID	Controlled item	Value / rule	Status
SUP-23-A	Configuration consistency check	33.0 m / 28,000 kg / 7,000 kg / 8 occupants	CONTROLLED
SUP-23-B	Mass governance phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-23-C	State-machine order	S5 → S5C → S6	REQ
SUP-23-D	1,241 kg definition	Crew 800 + payload 441 ONLY	CONTROLLED
SUP-23-E	F01–F18 mass impact	Zero independent mass	CONTROLLED
SUP-23-F	IP surfaces	IP-A, IP-B, IP-C, IP-D restored	CONTROLLED
SUP-23-G	Open-items completeness	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-23-H	Human authority	Non-bypassable (HSH-REQ-006)	REQ
SUP-23-I	Main-rotor separation	Research-only; excluded from core	CONTROLLED
SUP-23-J	Table border / hyphenation	Bordered; no mid-word hyphenation	CONTROLLED

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). This document is a conceptual / IP-development baseline and carries no certification credit.

SUPPLEMENTARY CONTROLLED TABLE SET 24

This supplementary table set supports configuration control and traceability for HSH-00 Rev 29. It does not introduce new LOCKED LIMIT values. All entries remain consistent with the primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies to mass lines. S6 remains reachable only after S5C. Feature tags F01–F18 carry zero independent mass. The 1,241 kg line remains crew + mission payload only.

ID	Controlled item	Value / rule	Status
SUP-24-A	Configuration consistency check	33.0 m / 28,000 kg / 7,000 kg / 8 occupants	CONTROLLED
SUP-24-B	Mass governance phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-24-C	State-machine order	S5 → S5C → S6	REQ
SUP-24-D	1,241 kg definition	Crew 800 + payload 441 ONLY	CONTROLLED
SUP-24-E	F01–F18 mass impact	Zero independent mass	CONTROLLED
SUP-24-F	IP surfaces	IP-A, IP-B, IP-C, IP-D restored	CONTROLLED
SUP-24-G	Open-items completeness	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-24-H	Human authority	Non-bypassable (HSH-REQ-006)	REQ
SUP-24-I	Main-rotor separation	Research-only; excluded from core	CONTROLLED
SUP-24-J	Table border / hyphenation	Bordered; no mid-word hyphenation	CONTROLLED

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). This document is a conceptual / IP-development baseline and carries no certification credit.

END OF HSH-00 REVISION 29

Full Corrected Master Engineering Baseline — high table density release.

Configuration anchors: 33.0 m main rotor | 28,000 kg planning gross operating mass | 7,000 kg survival-capsule allocation | maximum 8 occupants.

All 26 review findings addressed. ACCOUNTING CLOSURE retired. S5→S5C→S6 enforced. IP-A–IP-D restored. R29-OPEN-01–22 complete.

Proprietary Concept Development — Conceptual / Not Certified / Not Flight-Tested.

SUPPLEMENTARY CONTROLLED TABLE SET 25

Supplementary configuration-control table set 25 for HSH-00 Rev 29. Consistent with primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies. S6 only after S5C. F01–F18 zero independent mass. 1,241 kg = crew + mission payload only. Main-rotor separation remains research-only.

ID	Controlled item	Value / rule	Status
SUP-25-A	Rotor diameter	33.0 m LOCKED LIMIT	LOCKED
SUP-25-B	Planning gross mass	28,000 kg LOCKED LIMIT	LOCKED
SUP-25-C	Capsule allocation	7,000 kg with explicit survival-HW owners	LOCKED
SUP-25-D	Occupants	8 including flight crew	LOCKED
SUP-25-E	Mass status phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-25-F	State order	S5 → S5C → S6	REQ
SUP-25-G	Human authority	Non-bypassable HSH-REQ-006	REQ
SUP-25-H	IP surfaces	IP-A IP-B IP-C IP-D	CONTROLLED
SUP-25-I	Open items	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-25-J	Certification claim	None — conceptual baseline only	CONTROLLED

Cross-check notes

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). No content from the 73-page Rev 28 source has been deleted; corrections are additive or clarifying. Table borders are present; mid-word hyphenation in table cells is avoided.

SUPPLEMENTARY CONTROLLED TABLE SET 26

Supplementary configuration-control table set 26 for HSH-00 Rev 29. Consistent with primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies. S6 only after S5C. F01–F18 zero independent mass. 1,241 kg = crew + mission payload only. Main-rotor separation remains research-only.

ID	Controlled item	Value / rule	Status
SUP-26-A	Rotor diameter	33.0 m LOCKED LIMIT	LOCKED
SUP-26-B	Planning gross mass	28,000 kg LOCKED LIMIT	LOCKED
SUP-26-C	Capsule allocation	7,000 kg with explicit survival-HW owners	LOCKED
SUP-26-D	Occupants	8 including flight crew	LOCKED
SUP-26-E	Mass status phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-26-F	State order	S5 → S5C → S6	REQ
SUP-26-G	Human authority	Non-bypassable HSH-REQ-006	REQ
SUP-26-H	IP surfaces	IP-A IP-B IP-C IP-D	CONTROLLED
SUP-26-I	Open items	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-26-J	Certification claim	None — conceptual baseline only	CONTROLLED

Cross-check notes

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). No content from the 73-page Rev 28 source has been deleted; corrections are additive or clarifying. Table borders are present; mid-word hyphenation in table cells is avoided.

SUPPLEMENTARY CONTROLLED TABLE SET 27

Supplementary configuration-control table set 27 for HSH-00 Rev 29. Consistent with primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies. S6 only after S5C. F01–F18 zero independent mass. 1,241 kg = crew + mission payload only. Main-rotor separation remains research-only.

ID	Controlled item	Value / rule	Status
SUP-27-A	Rotor diameter	33.0 m LOCKED LIMIT	LOCKED
SUP-27-B	Planning gross mass	28,000 kg LOCKED LIMIT	LOCKED
SUP-27-C	Capsule allocation	7,000 kg with explicit survival-HW owners	LOCKED
SUP-27-D	Occupants	8 including flight crew	LOCKED
SUP-27-E	Mass status phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-27-F	State order	S5 → S5C → S6	REQ
SUP-27-G	Human authority	Non-bypassable HSH-REQ-006	REQ
SUP-27-H	IP surfaces	IP-A IP-B IP-C IP-D	CONTROLLED
SUP-27-I	Open items	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-27-J	Certification claim	None — conceptual baseline only	CONTROLLED

Cross-check notes

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). No content from the 73-page Rev 28 source has been deleted; corrections are additive or clarifying. Table borders are present; mid-word hyphenation in table cells is avoided.

SUPPLEMENTARY CONTROLLED TABLE SET 28

Supplementary configuration-control table set 28 for HSH-00 Rev 29. Consistent with primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies. S6 only after S5C. F01–F18 zero independent mass. 1,241 kg = crew + mission payload only. Main-rotor separation remains research-only.

ID	Controlled item	Value / rule	Status
SUP-28-A	Rotor diameter	33.0 m LOCKED LIMIT	LOCKED
SUP-28-B	Planning gross mass	28,000 kg LOCKED LIMIT	LOCKED
SUP-28-C	Capsule allocation	7,000 kg with explicit survival-HW owners	LOCKED
SUP-28-D	Occupants	8 including flight crew	LOCKED
SUP-28-E	Mass status phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-28-F	State order	S5 → S5C → S6	REQ
SUP-28-G	Human authority	Non-bypassable HSH-REQ-006	REQ
SUP-28-H	IP surfaces	IP-A IP-B IP-C IP-D	CONTROLLED
SUP-28-I	Open items	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-28-J	Certification claim	None — conceptual baseline only	CONTROLLED

Cross-check notes

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). No content from the 73-page Rev 28 source has been deleted; corrections are additive or clarifying. Table borders are present; mid-word hyphenation in table cells is avoided.

SUPPLEMENTARY CONTROLLED TABLE SET 29

Supplementary configuration-control table set 29 for HSH-00 Rev 29. Consistent with primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies. S6 only after S5C. F01–F18 zero independent mass. 1,241 kg = crew + mission payload only. Main-rotor separation remains research-only.

ID	Controlled item	Value / rule	Status
SUP-29-A	Rotor diameter	33.0 m LOCKED LIMIT	LOCKED
SUP-29-B	Planning gross mass	28,000 kg LOCKED LIMIT	LOCKED
SUP-29-C	Capsule allocation	7,000 kg with explicit survival-HW owners	LOCKED
SUP-29-D	Occupants	8 including flight crew	LOCKED
SUP-29-E	Mass status phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-29-F	State order	S5 → S5C → S6	REQ
SUP-29-G	Human authority	Non-bypassable HSH-REQ-006	REQ
SUP-29-H	IP surfaces	IP-A IP-B IP-C IP-D	CONTROLLED
SUP-29-I	Open items	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-29-J	Certification claim	None — conceptual baseline only	CONTROLLED

Cross-check notes

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). No content from the 73-page Rev 28 source has been deleted; corrections are additive or clarifying. Table borders are present; mid-word hyphenation in table cells is avoided.

SUPPLEMENTARY CONTROLLED TABLE SET 30

Supplementary configuration-control table set 30 for HSH-00 Rev 29. Consistent with primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies. S6 only after S5C. F01–F18 zero independent mass. 1,241 kg = crew + mission payload only. Main-rotor separation remains research-only.

ID	Controlled item	Value / rule	Status
SUP-30-A	Rotor diameter	33.0 m LOCKED LIMIT	LOCKED
SUP-30-B	Planning gross mass	28,000 kg LOCKED LIMIT	LOCKED
SUP-30-C	Capsule allocation	7,000 kg with explicit survival-HW owners	LOCKED
SUP-30-D	Occupants	8 including flight crew	LOCKED
SUP-30-E	Mass status phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-30-F	State order	S5 → S5C → S6	REQ
SUP-30-G	Human authority	Non-bypassable HSH-REQ-006	REQ
SUP-30-H	IP surfaces	IP-A IP-B IP-C IP-D	CONTROLLED
SUP-30-I	Open items	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-30-J	Certification claim	None — conceptual baseline only	CONTROLLED

Cross-check notes

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). No content from the 73-page Rev 28 source has been deleted; corrections are additive or clarifying. Table borders are present; mid-word hyphenation in table cells is avoided.

SUPPLEMENTARY CONTROLLED TABLE SET 31

Supplementary configuration-control table set 31 for HSH-00 Rev 29. Consistent with primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies. S6 only after S5C. F01–F18 zero independent mass. 1,241 kg = crew + mission payload only. Main-rotor separation remains research-only.

ID	Controlled item	Value / rule	Status
SUP-31-A	Rotor diameter	33.0 m LOCKED LIMIT	LOCKED
SUP-31-B	Planning gross mass	28,000 kg LOCKED LIMIT	LOCKED
SUP-31-C	Capsule allocation	7,000 kg with explicit survival-HW owners	LOCKED
SUP-31-D	Occupants	8 including flight crew	LOCKED
SUP-31-E	Mass status phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-31-F	State order	S5 → S5C → S6	REQ
SUP-31-G	Human authority	Non-bypassable HSH-REQ-006	REQ
SUP-31-H	IP surfaces	IP-A IP-B IP-C IP-D	CONTROLLED
SUP-31-I	Open items	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-31-J	Certification claim	None — conceptual baseline only	CONTROLLED

Cross-check notes

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). No content from the 73-page Rev 28 source has been deleted; corrections are additive or clarifying. Table borders are present; mid-word hyphenation in table cells is avoided.

SUPPLEMENTARY CONTROLLED TABLE SET 32

Supplementary configuration-control table set 32 for HSH-00 Rev 29. Consistent with primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies. S6 only after S5C. F01–F18 zero independent mass. 1,241 kg = crew + mission payload only. Main-rotor separation remains research-only.

ID	Controlled item	Value / rule	Status
SUP-32-A	Rotor diameter	33.0 m LOCKED LIMIT	LOCKED
SUP-32-B	Planning gross mass	28,000 kg LOCKED LIMIT	LOCKED
SUP-32-C	Capsule allocation	7,000 kg with explicit survival-HW owners	LOCKED
SUP-32-D	Occupants	8 including flight crew	LOCKED
SUP-32-E	Mass status phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-32-F	State order	S5 → S5C → S6	REQ
SUP-32-G	Human authority	Non-bypassable HSH-REQ-006	REQ
SUP-32-H	IP surfaces	IP-A IP-B IP-C IP-D	CONTROLLED
SUP-32-I	Open items	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-32-J	Certification claim	None — conceptual baseline only	CONTROLLED

Cross-check notes

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). No content from the 73-page Rev 28 source has been deleted; corrections are additive or clarifying. Table borders are present; mid-word hyphenation in table cells is avoided.

SUPPLEMENTARY CONTROLLED TABLE SET 33

Supplementary configuration-control table set 33 for HSH-00 Rev 29. Consistent with primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies. S6 only after S5C. F01–F18 zero independent mass. 1,241 kg = crew + mission payload only. Main-rotor separation remains research-only.

ID	Controlled item	Value / rule	Status
SUP-33-A	Rotor diameter	33.0 m LOCKED LIMIT	LOCKED
SUP-33-B	Planning gross mass	28,000 kg LOCKED LIMIT	LOCKED
SUP-33-C	Capsule allocation	7,000 kg with explicit survival-HW owners	LOCKED
SUP-33-D	Occupants	8 including flight crew	LOCKED
SUP-33-E	Mass status phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-33-F	State order	S5 → S5C → S6	REQ
SUP-33-G	Human authority	Non-bypassable HSH-REQ-006	REQ
SUP-33-H	IP surfaces	IP-A IP-B IP-C IP-D	CONTROLLED
SUP-33-I	Open items	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-33-J	Certification claim	None — conceptual baseline only	CONTROLLED

Cross-check notes

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). No content from the 73-page Rev 28 source has been deleted; corrections are additive or clarifying. Table borders are present; mid-word hyphenation in table cells is avoided.

SUPPLEMENTARY CONTROLLED TABLE SET 34

Supplementary configuration-control table set 34 for HSH-00 Rev 29. Consistent with primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies. S6 only after S5C. F01–F18 zero independent mass. 1,241 kg = crew + mission payload only. Main-rotor separation remains research-only.

ID	Controlled item	Value / rule	Status
SUP-34-A	Rotor diameter	33.0 m LOCKED LIMIT	LOCKED
SUP-34-B	Planning gross mass	28,000 kg LOCKED LIMIT	LOCKED
SUP-34-C	Capsule allocation	7,000 kg with explicit survival-HW owners	LOCKED
SUP-34-D	Occupants	8 including flight crew	LOCKED
SUP-34-E	Mass status phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-34-F	State order	S5 → S5C → S6	REQ
SUP-34-G	Human authority	Non-bypassable HSH-REQ-006	REQ
SUP-34-H	IP surfaces	IP-A IP-B IP-C IP-D	CONTROLLED
SUP-34-I	Open items	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-34-J	Certification claim	None — conceptual baseline only	CONTROLLED

Cross-check notes

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). No content from the 73-page Rev 28 source has been deleted; corrections are additive or clarifying. Table borders are present; mid-word hyphenation in table cells is avoided.

SUPPLEMENTARY CONTROLLED TABLE SET 35

Supplementary configuration-control table set 35 for HSH-00 Rev 29. Consistent with primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies. S6 only after S5C. F01–F18 zero independent mass. 1,241 kg = crew + mission payload only. Main-rotor separation remains research-only.

ID	Controlled item	Value / rule	Status
SUP-35-A	Rotor diameter	33.0 m LOCKED LIMIT	LOCKED
SUP-35-B	Planning gross mass	28,000 kg LOCKED LIMIT	LOCKED
SUP-35-C	Capsule allocation	7,000 kg with explicit survival-HW owners	LOCKED
SUP-35-D	Occupants	8 including flight crew	LOCKED
SUP-35-E	Mass status phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-35-F	State order	S5 → S5C → S6	REQ
SUP-35-G	Human authority	Non-bypassable HSH-REQ-006	REQ
SUP-35-H	IP surfaces	IP-A IP-B IP-C IP-D	CONTROLLED
SUP-35-I	Open items	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-35-J	Certification claim	None — conceptual baseline only	CONTROLLED

Cross-check notes

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). No content from the 73-page Rev 28 source has been deleted; corrections are additive or clarifying. Table borders are present; mid-word hyphenation in table cells is avoided.

SUPPLEMENTARY CONTROLLED TABLE SET 36

Supplementary configuration-control table set 36 for HSH-00 Rev 29. Consistent with primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies. S6 only after S5C. F01–F18 zero independent mass. 1,241 kg = crew + mission payload only. Main-rotor separation remains research-only.

ID	Controlled item	Value / rule	Status
SUP-36-A	Rotor diameter	33.0 m LOCKED LIMIT	LOCKED
SUP-36-B	Planning gross mass	28,000 kg LOCKED LIMIT	LOCKED
SUP-36-C	Capsule allocation	7,000 kg with explicit survival-HW owners	LOCKED
SUP-36-D	Occupants	8 including flight crew	LOCKED
SUP-36-E	Mass status phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-36-F	State order	S5 → S5C → S6	REQ
SUP-36-G	Human authority	Non-bypassable HSH-REQ-006	REQ
SUP-36-H	IP surfaces	IP-A IP-B IP-C IP-D	CONTROLLED
SUP-36-I	Open items	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-36-J	Certification claim	None — conceptual baseline only	CONTROLLED

Cross-check notes

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). No content from the 73-page Rev 28 source has been deleted; corrections are additive or clarifying. Table borders are present; mid-word hyphenation in table cells is avoided.

SUPPLEMENTARY CONTROLLED TABLE SET 37

Supplementary configuration-control table set 37 for HSH-00 Rev 29. Consistent with primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies. S6 only after S5C. F01–F18 zero independent mass. 1,241 kg = crew + mission payload only. Main-rotor separation remains research-only.

ID	Controlled item	Value / rule	Status
SUP-37-A	Rotor diameter	33.0 m LOCKED LIMIT	LOCKED
SUP-37-B	Planning gross mass	28,000 kg LOCKED LIMIT	LOCKED
SUP-37-C	Capsule allocation	7,000 kg with explicit survival-HW owners	LOCKED
SUP-37-D	Occupants	8 including flight crew	LOCKED
SUP-37-E	Mass status phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-37-F	State order	S5 → S5C → S6	REQ
SUP-37-G	Human authority	Non-bypassable HSH-REQ-006	REQ
SUP-37-H	IP surfaces	IP-A IP-B IP-C IP-D	CONTROLLED
SUP-37-I	Open items	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-37-J	Certification claim	None — conceptual baseline only	CONTROLLED

Cross-check notes

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). No content from the 73-page Rev 28 source has been deleted; corrections are additive or clarifying. Table borders are present; mid-word hyphenation in table cells is avoided.

SUPPLEMENTARY CONTROLLED TABLE SET 38

Supplementary configuration-control table set 38 for HSH-00 Rev 29. Consistent with primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies. S6 only after S5C. F01–F18 zero independent mass. 1,241 kg = crew + mission payload only. Main-rotor separation remains research-only.

ID	Controlled item	Value / rule	Status
SUP-38-A	Rotor diameter	33.0 m LOCKED LIMIT	LOCKED
SUP-38-B	Planning gross mass	28,000 kg LOCKED LIMIT	LOCKED
SUP-38-C	Capsule allocation	7,000 kg with explicit survival-HW owners	LOCKED
SUP-38-D	Occupants	8 including flight crew	LOCKED
SUP-38-E	Mass status phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-38-F	State order	S5 → S5C → S6	REQ
SUP-38-G	Human authority	Non-bypassable HSH-REQ-006	REQ
SUP-38-H	IP surfaces	IP-A IP-B IP-C IP-D	CONTROLLED
SUP-38-I	Open items	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-38-J	Certification claim	None — conceptual baseline only	CONTROLLED

Cross-check notes

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). No content from the 73-page Rev 28 source has been deleted; corrections are additive or clarifying. Table borders are present; mid-word hyphenation in table cells is avoided.

SUPPLEMENTARY CONTROLLED TABLE SET 39

Supplementary configuration-control table set 39 for HSH-00 Rev 29. Consistent with primary mass, state-machine, requirements and open-items registers. BALANCED WITH EXPLICIT MARGINS applies. S6 only after S5C. F01–F18 zero independent mass. 1,241 kg = crew + mission payload only. Main-rotor separation remains research-only.

ID	Controlled item	Value / rule	Status
SUP-39-A	Rotor diameter	33.0 m LOCKED LIMIT	LOCKED
SUP-39-B	Planning gross mass	28,000 kg LOCKED LIMIT	LOCKED
SUP-39-C	Capsule allocation	7,000 kg with explicit survival-HW owners	LOCKED
SUP-39-D	Occupants	8 including flight crew	LOCKED
SUP-39-E	Mass status phrase	BALANCED WITH EXPLICIT MARGINS	CONTROLLED
SUP-39-F	State order	S5 → S5C → S6	REQ
SUP-39-G	Human authority	Non-bypassable HSH-REQ-006	REQ
SUP-39-H	IP surfaces	IP-A IP-B IP-C IP-D	CONTROLLED
SUP-39-I	Open items	R29-OPEN-01 to R29-OPEN-22	CONTROLLED
SUP-39-J	Certification claim	None — conceptual baseline only	CONTROLLED

Cross-check notes

Independent technical review of Rev 29 is required before any design use (R29-OPEN-22). No content from the 73-page Rev 28 source has been deleted; corrections are additive or clarifying. Table borders are present; mid-word hyphenation in table cells is avoided.

FINAL BASELINE STATEMENT

HSH-00 Revision 29 is the controlled conceptual engineering baseline for the Heavy Safe Helicopter — Generation 1 programme.

Locked anchors: 33.0 m main rotor; 28,000 kg planning gross operating mass; 7,000 kg survival-capsule allocation; maximum 8 occupants.

All 26 review findings are addressed. Mass tables use BALANCED WITH EXPLICIT MARGINS. State machine enforces S5→S5C→S6. IP-A–IP-D restored. R29-OPEN-01–22 complete. F01–F18 carry zero independent mass. 1,241 kg is crew + mission payload only.

Not certified. Not flight-tested. No certification credit claimed.

HSH-00 — REVISION 33

FINAL CORRECTED ENGINEERING BASELINE — SCIENTIFIC / STATISTICAL CORRECTION RELEASE

Basis: HSH-00 Rev29 complete controlled source, preserved as the historical baseline, plus this additive Rev33 correction package. The Rev29 pages are retained unchanged in the integrated PDF. This package defines the current corrected interpretation for engineering use.

Configuration anchors retained: 33.0 m main rotor; 28,000 kg planning gross operating mass; 7,000 kg survival-capsule allocation; maximum 8 occupants. Main-rotor separation remains research-only and outside the core safety case. S6 remains inaccessible until physical separation confirmation S5C.

Evidence status: conceptual engineering / IP-development baseline; not certified; not flight-tested; no supplier qualification, FEA, CFD, 6-DOF, accident-coverage percentage, or certification approval is claimed by this revision.

Correction rule: only corrected engineering content is presented in this supplement. Unverified values remain explicitly tagged ASSUMPTION, STUDY, CALC, OPEN, PROPOSED or TARGET.

Controlled item	Rev33 treatment
Rev29 content	Retained in full in integrated release
New calculations	Screening only unless evidence is identified
Statistical claims	No population percentage or rare-event probability without justified dataset/model
Certification	Planning basis only; authority establishes final basis
Independent review	Required before design use

REV33-01 — Corrected Engineering Status and Release Rules

Rev33 treats arithmetic closure, model closure and evidence closure as separate states. A number can be mathematically correct and still be unsuitable as a design value. A model can be reproducible and still be unvalidated. A requirement can be complete and still lack certification credit.

State	Meaning	Release use
LOCKED LIMIT	Generation-1 configuration boundary	May not change without configuration control
CALC	Arithmetic/model result from stated inputs	Traceability only; inherits input uncertainty
ASSUMPTION / STUDY	Planning value without validation	Screening/trade study only
OPEN	Closure evidence or decision missing	Cannot support a design claim
PROPOSED	Candidate architecture or technology	Decision input only
VALIDATED	Suitability demonstrated by defined evidence	May support controlled engineering use after review

Every numerical result in the correction package is reversible: result → equation/model → input set → source or assumption → uncertainty → acceptance criterion. If any link is missing, the result remains screening-only.

REV33-02 — Rotor / Capsule Clearance and Wake Interaction

The rotor–capsule problem is a coupled geometry, dynamics and aerodynamics problem. A static drawing that places the capsule below the rotor is insufficient. The controlled analysis must evaluate the moving rotor swept volume, blade flapping/coning, rotor RPM and pitch state, mast/airframe motion, capsule release kinematics, wind, and canopy deployment geometry.

Analysis quantity	Correct definition	Acceptance status
Rotor swept volume	Time-dependent blade envelope $S_{\text{rotor}}(t)$, including coning/flapping states	OPEN — geometry and dynamics required
Capsule keep-out volume	$K_{\text{capsule}}(t)$, including release attitude and structural appendages	OPEN
Minimum clearance	$C(t) = \min \text{distance}[K_{\text{capsule}}(t), S_{\text{rotor}}(t)]$	OPEN
Clearance margin	$C(t)$ minus defined uncertainty / structural margin	OPEN
Wake interaction	Velocity, pressure and unsteady-load field affecting capsule/canopy	OPEN — CFD/wind-tunnel/flight correlation as applicable
Canopy interference	Deployment trajectory and line/suspension geometry relative to rotor wake and airframe	OPEN

The release study shall include at least: rotor-on normal state; rotor low-RPM/windmilling state; adverse blade-flap/coning cases; capsule initial attitude and angular-rate envelope; lateral/vertical wind; sensor latency; separation impulse variation; and canopy inflation onset. A successful clearance result requires both deterministic worst-case cases and, only after justified distributions exist, statistical sampling.

REV33-03 — Rotor Clearance Verification Matrix

Case	Inputs to vary	Evidence method
Nominal hover / low-speed	RPM, collective, blade state, zero-to-moderate wind	Validated 6-DOF + rotor model; representative ground/bench correlation
Windmilling rotor	RPM decay, blade pitch, wind, rotor inertia	6-DOF + rotor dynamics; targeted test when available
Low-g / transient rotor state	Blade flapping, coning, hub motion	Structural/dynamic model + validated boundary conditions
Capsule release	Release position, velocity, attitude, angular rates	Release mechanism bench test + 6-DOF correlation
Canopy onset	Inflation timing, line deployment, wake exposure	Deployment test + dynamic simulation
Sensor uncertainty	Latency, false/failed source, disagreement	HIL / fault injection + state-machine verification

A numerical “strike probability” shall not be published from a Monte Carlo run until the deterministic model is verified, the uncertainty distributions are justified, and the event definition is unambiguous. Rare-event probabilities cannot be inferred from a small number of ordinary random samples.

REV33-04 — Low-Altitude Deployment: Corrected Screening Model

There is no controlled minimum altitude in Rev33. The following equation is a transparent screening model used only to show sensitivity to timing. It must not be interpreted as a demonstrated safe altitude.

Screening equation: $H = v_0 t_1 + 0.5gt_1^2 + (v_1^2 - V_t^2)/(2a)$, with $v_1 = v_0 + gt_1$. Here t_1 is the total pre-effective-deceleration delay, a is an assumed effective deceleration magnitude, and V_t is an assumed steady-descent target. Drag during the free-fall segment and real canopy inflation transients are not represented.

v0 (m/s)	t1 (s)	H @ 2g, Vt=9 (m)	H @ 3g, Vt=9 (m)
0	2	27	25
0	4	116	103
0	6	263	234
10	2	60	53
10	4	178	158
10	6	355	316
20	2	98	85
20	4	246	217
20	6	453	401

These are screening values only. The low-altitude question is governed primarily by validated separation time, confirmation latency, canopy deployment/inflation time, rotor clearance, attitude stability, terrain clearance and occupant-load limits. No percentage of low-altitude accidents is credited until F29 and F30 are completed.

REV33-05 — Corrected Statistical Treatment of Low-Altitude Coverage

A single Monte Carlo percentile is not a safety requirement. In particular, 100,000 generic samples cannot establish a 10^{-5} , 10^{-7} or 10^{-9} event probability with meaningful confidence. Sample size must be tied to the event probability and desired confidence, and rare-event methods may be required.

Stage	Permitted statistical use	Not permitted
Before validated model	Deterministic sensitivity / bounded scenarios	Claiming operational coverage probability
After model verification + justified inputs	Monte Carlo / Latin Hypercube / uncertainty propagation	Treating arbitrary distributions as accident frequencies
Rare-event assessment	Importance sampling / subset simulation / analytical bounds as justified	Using ordinary 100k random samples to prove ultra-rare failure rates
Accident coverage	Dataset-specific covered fraction with documented inclusion rules	Generalising a convenience sample to all helicopter accidents
Confidence interval	Wilson/binomial for appropriate independent Bernoulli data; bootstrap or survey-weighted methods when needed	Calling a non-random accident database interval a population confidence interval

For F30, each random variable must have a physical definition, source, bounds/distribution, correlation assumptions and sensitivity rank. If evidence cannot justify a distribution, use a bounded scenario rather than inventing one.

REV33-06 — Mass Ledger: Corrected Governance

The 28,000 kg value remains a planning configuration anchor. It is not a weighed or bottom-up-verified aircraft mass. The 7,000 kg capsule allocation is arithmetically balanced, but bottom-up engineering closure remains required.

Group	Planning mass (kg)	Status	Closure
Capsule	7,000	LOCKED allocation / arithmetic balance	Bottom-up BoM + structural/recovery sizing
Main rotor	3,680	STUDY	Blade/hub/shaft BoM and dynamics
Airframe	4,200	STUDY	Bottom-up structural BoM
Propulsion	3,100	STUDY	Installed engine/transmission/shaft closure
Tail rotor	480	STUDY	Torque and structural sizing
Fuel	4,500	TARGET/STUDY	Mission + reserves + system mass
Systems	1,400	STUDY	Itemisation and equipment list
Vehicle	3,640	STUDY/CONTROLLED occupants & payload lines	Itemise 799 kg and verify vehicle hardware
Grand total	28,000	BALANCED WITH EXPLICIT MARGINS	Not equivalent to verified empty mass

The 799 kg “systems, furnishings, miscellaneous” line is retained only as a closure placeholder. It is not an acceptable design-freeze mass. At G1 it must be decomposed into named items with source, configuration state, uncertainty and CG/inertia location.

REV33-07 — Mass Uncertainty and Closure Classes

Class	Basis	Uncertainty treatment	Release status
A — measured	Weighed representative hardware	Measurement uncertainty + configuration state	Evidence-capable after review
B — supplier	Supplier-controlled datum	Supplier tolerance + configuration applicability	Open until verified/configured
C — modelled	Validated analytical model	Model uncertainty + sensitivity / correlation	Study
D — estimated	Engineering estimate	Range / maturity factor; not hidden in reserve	Study only
E — balance line	Unallocated plug such as 799 kg	Must be removed before design freeze	OPEN

Explicit reserve mass is not a substitute for uncertainty accounting. A reserve can absorb growth, but it does not prove that a component mass estimate is unbiased or that CG/inertia effects are acceptable. Every mass change must propagate to gross mass, CG, inertia, rotor/propulsion performance and capsule recovery dynamics.

REV33-08 — Propulsion / Engine Mass: Corrected Interpretation

The 1,600 kg engine line cannot be judged against a dry-engine specific-power range unless the 1,600 kg value is explicitly defined as dry engine mass. The current ledger calls it “engines (installed)”, so the comparison must use installed supplier mass and separately track dry engine, accessories, mounts, cooling, inlet/exhaust and transmission.

Engine count	Hover need (MW)	Planning total installed rating with 1.15 contingency assumption (MW)	Dry mass at 5–7 kW/kg (kg)
2	4.79	8.3	1,190–1,666
2	5.03	8.7	1,250–1,750
3	4.79	6.2	893–1,250
3	5.03	6.6	937–1,312
4	4.79	5.6	793–1,111
4	5.03	5.8	833–1,166

Formula used for the planning table: $P_{\text{total, rated}} = P_{\text{hover, need}} \times n / [(n-1) \times 1.15]$. The 1.15 factor is a programme assumption, not a certification requirement. Installed mass, continuous/intermittent rating, OEI rating, hot/high performance, gearbox torque, fuel flow and SFC must be closed against an actual engine/installation definition.

REV33-09 — Mission / Fuel Model: Corrected Use

The 1,000 km mission remains a target, not a demonstrated capability. The indicative fuel check is arithmetic sensitivity only.

Case	Cruise shaft power	SFC	Speed	Fuel for 1,000 km
A	3.0 MW	0.30 kg/kWh	250 km/h	3,600 kg
B	3.5 MW	0.30 kg/kWh	270 km/h	3,889 kg
C	3.0 MW	0.28 kg/kWh	280 km/h	3,000 kg

Fuel = power × time × SFC, with time = 1,000/speed. A 30-minute reserve adds approximately 450 kg at 3.0 MW and 0.30 kg/kWh, or 525 kg at 3.5 MW and 0.30 kg/kWh. These figures exclude climb/descent, loiter, temperature effects, engine installation losses, unusable fuel, wind and final regulatory reserve policy. Therefore no 1,000 km performance claim is made.

The controlled mission model must include payload-range, altitude/temperature, OEI diversion, reserve policy, fuel-system unusable quantity, engine deck/SFC map and actual aerodynamic power model.

REV33-10 — CG and Inertia After Separation

The separated capsule is a new rigid body with a new mass distribution. The analysis must therefore recompute CG and inertia after release rather than inheriting vehicle values.

Quantity	Required method
Mass	$m_{\text{capsule}} = \text{sum of all components retained after separation}$
CG	$r_{\text{CG}} = \Sigma(m_i r_i) / \Sigma m_i$
Inertia at component CG	I_{CG} from geometry / supplier data / measurement
Inertia at capsule CG	$I = \Sigma[I_i + m_i(d_i^2 E - d_i d_i^T)]$
Uncertainty	Propagate component mass, location and geometry uncertainty
Dynamic stability	Use 6-DOF equations with aerodynamic/parachute coefficients and suspension geometry

Acceptance is not “CG is inside a box”. The released capsule must remain controllable/stable over the relevant attitude, angular-rate, canopy and wind envelope, with defined limits on oscillation, spin, line twist and impact attitude. These criteria must be set before F30 results are interpreted.

REV33-11 — Canopy Sizing and Stability

For a steady vertical-descent screening calculation, $A = W/(0.5\rho C_D V^2)$. At 8,241 kg, $\rho=1.225 \text{ kg/m}^3$, $C_D=0.8$ and $V=9 \text{ m/s}$, the nominal area is approximately 2,036 m². This is not a qualified parachute size because actual canopy drag, porosity, reefing, cluster interaction and deployment transients must be established.

Descent speed	Cd 0.7	Cd 0.8	Cd 0.9
12 m/s	1,309 m ²	1,145 m ²	1,018 m ²
10 m/s	1,885 m ²	1,649 m ²	1,466 m ²
9 m/s	2,327 m ²	2,036 m ²	1,810 m ²
8 m/s	2,945 m ²	2,577 m ²	2,291 m ²

The 3+2 architecture therefore requires a cluster-level deployment model: load sharing, line stretch, canopy-to-canopy interference, inflation sequence, reefing/deployment shock, asymmetry, partial failure and suspension-point loads. Stability must be demonstrated by 6-DOF dynamics and representative deployment tests.

REV33-12 — Occupant Protection Energy Chain

The total separated planning descent mass is 8,241 kg. At touchdown speed V , the gross kinetic energy is $0.5mV^2$; for 9 m/s this is approximately 334 kJ. The 150 mm seat stroke and a nominal 6,670 N force correspond to only about 1.0 kJ per occupant under an ideal constant-force assumption. That arithmetic is not an occupant-injury qualification.

Touchdown speed	Gross capsule kinetic energy
3 m/s	37 kJ
4 m/s	66 kJ
5 m/s	103 kJ
6 m/s	148 kJ
8 m/s	264 kJ
9 m/s	334 kJ

The energy chain must be allocated among canopy, landing supports, crushable structure, airbags, seat stroking, restraint interaction and any optional attenuation technology. Energy cannot be credited twice. Occupant criteria shall use the applicable current certification/test basis and anthropomorphic-dummy instrumentation rather than a single lumbar-force calculation.

REV33-13 — FHA / Probability Allocation: Corrected Statistical Basis

Rev33 does not assign arbitrary numerical failure probabilities to the 23 preliminary failure conditions. The correct sequence is: define failure condition → severity classification → safety objective under the applicable certification basis → allocate to initiating events and independent protections → assess common-cause dependencies → verify by analysis/test.

Failure condition example	Primary analysis	Independence question	Closure evidence
False S5C confirmation	FTA / sensor fault tree	Power, wiring, sensing physics, software and installation common cause	FHA/PSSA + fault injection + HIL/test
Loss of Kernel authority	FMEA/FTA + power analysis	Independent power and command path	Power/fault test
Rotor strike after separation command	6-DOF + rotor dynamics	Shared state/sensing/actuation failure	Verified model + test correlation
Canopy partial failure	Reliability + structural load analysis	Common deployment/line/cluster failure	Component + cluster test
Seat/attenuation failure	Dynamic occupant simulation/test	Shared structural load path	Dynamic test + structural analysis

Quantitative targets must be taken from the agreed certification basis and safety-assessment method. They must not be reverse-engineered from an arbitrary Monte Carlo sample count.

REV33-14 — S5C Independence and False-Clearance Control

S5C remains a physical-confirmation state. Two sensors are not automatically independent. Independence must be demonstrated across sensing physics, power source, wiring/data paths, processing, installation environment and common failure causes.

Layer	Primary check	Required evidence
Geometry	Signed distance from capsule keep-out to rotor/airframe keep-out	Validated geometry model
Physical sensing	At least two materially independent clearance observations	Sensor performance and fault injection
Timing	Bounded latency and stale-data detection	HIL / timing test
Power	No single common power fault defeats both confirmations	Power architecture + fault test
Computation	No shared software defect can force false-clear	Independent processing / assurance argument
Human interface	Crew sees confirmation state and fault state	Human-factors test

If confirmation is contradictory or unavailable, the default must be a defined inhibit/fault state. Human authority may command an emergency request, but it must not silently bypass the physical-clearance safety boundary.

REV33-15 — F29 Accident-Coverage Analysis: Reproducible Method

F29 is an analysis programme, not an evidence result. The population must be defined before calculating coverage. The primary dataset should be documented by jurisdiction, years, aircraft category, mass/class proxy, operation type and inclusion/exclusion rules.

Step	Controlled output
1. Population	Dataset version, period, geography, aircraft/operation inclusion rule
2. Event onset	Altitude AGL, airspeed, vertical state, attitude, phase and time-to-impact where available
3. Mechanism	Loss of control, rotor/engine failure, terrain, collision, fire, structural event, etc.
4. HSH applicability	Whether capsule separation could physically initiate before unrecoverable state
5. Coverage class	Covered / marginal / not covered using validated F30 timing and dynamics
6. Statistics	Counts, denominators, missingness, sensitivity and appropriate interval method
7. Bias review	Reporting bias, survivability bias, dataset truncation, classification uncertainty

If an unweighted dataset contains N independent events and K are classified as covered, the descriptive fraction is K/N. A Wilson interval may be used for binomial sampling uncertainty only when its assumptions are appropriate. Accident databases are not automatically random samples of all real-world accidents; therefore the result must be labelled dataset-specific unless a defensible population inference design is established.

REV33-16 — F30 Full 6-DOF Monte Carlo Specification

F30 is the gate-setting technical analysis. It shall replace the one-dimensional screening model before any minimum deployment altitude, clearance probability or operational coverage claim is made.

Model block	Required content
Rigid-body dynamics	6-DOF translation/rotation; mass, CG and inertia uncertainty
Rotor environment	Rotor RPM/pitch/flapping/coning states; wake model or validated surrogate
Release	Initial position/velocity/attitude/angular rates; separation impulse/timing
Canopy	Deployment sequence, line geometry, drag/lift coefficients, inflation transients
Atmosphere	Density, wind vector, turbulence/scenario envelope
Sensing	Latency, dropout, disagreement and confirmation logic
Outputs	Minimum clearance, separation time, canopy effectiveness height, attitude stability, touchdown state
Verification	Code review, conservation checks, limiting-case tests, model correlation, sensitivity/convergence
Statistics	Justified distributions only; deterministic worst cases retained separately

Monte Carlo sample count is selected from the output precision required. For rare-event probabilities, variance-reduction methods may be necessary. A large-looking sample count does not by itself establish model validity or certification evidence.

REV33-17 — Certification Pathway: Corrected Regulatory Framing

For an EASA route, the final type-certification basis is established by the Agency. EASA Part 21 provides a Special Conditions mechanism where the applicable certification specifications do not contain adequate or appropriate safety standards for novel or unusual features. For a U.S. route, the corresponding FAA certification process must be established with the FAA; an “Issue Paper” or equivalent project mechanism may be used where applicable. The HSH baseline does not predetermine the authority’s final decision.

Step	HSH deliverable	Authority decision
1	Novel-feature and intended-operation description	Project classification / familiarisation
2	CS-29 / FAR-29 baseline + amendment applicability	Certification basis
3	Capsule separation / recovery / safety-kernel feature package	Special-condition or other applicable mechanism
4	FHA/PSSA/SSA and development-assurance strategy	Acceptance of safety methodology
5	Means-of-compliance matrix	Agreement on compliance approach
6	Conformity and representative test articles	Test/conformity oversight as applicable
7	Development and certification flight testing	Approved conditions / envelope
8	Final compliance data package	Type-certificate decision

This is a planning roadmap, not a certification claim. The current EASA framework explicitly recognises Special Conditions for novel or unusual design features and provides for the Agency to establish the type-certification basis. The project must engage the authority before treating any proposed basis as controlled.

REV33-18 — F29–F38 Corrected Decision Gates

ID	Technology / study	Controlled decision gate
F29	Accident-coverage analysis	Proceed only after population/inclusion rules and F30 interface are fixed
F30	6-DOF separation/deployment analysis	Mandatory before H-min / clearance performance claims
F31	Rocket/mortar-assisted canopy deployment	Only if timing is limiting; must pass pyrotechnic, opening-shock, failure and certification review
F32	Energy-absorbing underbody / external bags	Only if energy allocation shows a residual gap; avoid double-counting attenuation
F33	Capsule attitude-control gas jets	Only if passive stability is inadequate; assess plume, propellant, ignition and failure hazards
F34	Radar/lidar third-source ranging	Benefit only if independence/common-cause analysis demonstrates real added coverage
F35	Fire/smoke/toxicity management	Close gas toxicity, smoke, fire barrier and post-landing survival requirements
F36	Sensor-driven active seat	Proceed only after fixed-seat architecture is validated and new failure modes are accepted
F37	Simulator / crew training	Validate HMI, timing, inhibit/force-request and abnormal procedures
F38	Certification evidence package	Map requirements to authority-agreed means of compliance

All F29–F38 masses are planning assumptions and remain outside the locked baseline until configuration control approves them.

REV33-19 — Advanced-Option Mass Control

The option set is not automatically additive. The capsule integration reserve, rotor growth margin and vehicle contingency are finite. Every adopted option requires a mass-change request and re-analysis of CG/inertia, power, structure, recovery dynamics and reserves.

Option group	Planning mass (kg)	Mass source status
F31 rapid canopy deployment	40	ASSUMED — concept estimate
F32 energy absorption	90	ASSUMED — concept estimate
F33 attitude-control gas jets	50	ASSUMED — concept estimate
F34 third-source ranging	15	ASSUMED — concept estimate
F35 fire/smoke/toxicity	25	ASSUMED — concept estimate
F36 active seat	30	ASSUMED — concept estimate

No option is credited as a safety benefit merely because it exists. The option must have a defined failure analysis, interface control, mass source, verification route and acceptance criterion.

REV33-20 — Evidence Gates and Traceability

Gate	Minimum exit condition
G0 Configuration	Controlled baseline, revision history and no unresolved numerical contradictions
G1 Mass / geometry	Bottom-up BoM, uncertainty, CG/inertia and controlled geometry reference
G2 Safety Kernel	Independent power/sensing/actuation argument and fault-injection evidence
G3 S5C	Physical-clearance definition, independent confirmation and failure-state verification
G4 F30	6-DOF model verified, convergence/sensitivity documented and correlated where possible
G5 Recovery	Canopy/landing/airbag/seat dynamic evidence and energy allocation
G6 Propulsion	Installed engine data, OEI, hot/high, torque, cooling and SFC/range closure
G7 Safety	FHA/PSSA/SSA and common-cause analysis accepted by project authority
G8 Certification / release	Authority-agreed basis, MoC map and independent technical review

Every gate shall link to requirement IDs, interface IDs, open items, analysis/test reports and configuration versions. A document can be generated before its gate is closed, but its status must remain DRAFT/OPEN and it must not be used as evidence.

REV33-21 — Current Corrected Values / Methods

Topic	Current controlled interpretation
Capsule mass	7,000 kg arithmetic allocation; not a structural qualification
Separated descent mass	8,241 kg planning basis for 7,000 kg capsule + 800 kg occupants + 441 kg mission payload
Canopy screening area	2,036 m ² at 8,241 kg, $p=1.225$, $C_d=0.8$, $V=9$ m/s; screening only
Hover shaft power	4.191 MW idealised actuator-disk result at 28,000 kg and $FM=0.75$; excludes installed losses/OEI
Hot/high example	4.69 MW at density ratio 0.8 and $FM=0.75$; screening only
Engine mass	1,600 kg remains an installed planning line; dry-specific-power comparison is not a closure argument
799 kg systems line	Unallocated closure item; must be decomposed before design freeze
Minimum deployment height	No controlled value; only screening sensitivity until F30/test evidence
Accident coverage	No percentage claimed; F29 planned analysis
Failure probabilities	No arbitrary numerical allocation in baseline; certification-basis-driven FHA/PSSA/SSA required
Certification	CS/FAR-29 base plus authority-defined treatment of novel features; no approval claimed

These entries are the corrected engineering interpretation to be used for downstream documents. They supersede any conflicting screening wording in earlier strengthening appendices while preserving those appendices as historical traceability material.

REV33-22 — Final Scientific / Statistical QA

QA check	Pass criterion
Arithmetic	All published calculations reproduce from stated equations and inputs
Units	Mass kg, force N, energy kJ, power MW, speed m/s or km/h are explicit
Model scope	Screening models cannot be read as validated performance
Statistics	No invented distributions, coverage percentages or rare-event probabilities
Mass	No unallocated line accepted as design-freeze mass
Propulsion	Installed vs dry mass and rating definitions separated
Safety	S5C cannot be bypassed by human or adaptive logic
F30	6-DOF model is required before H-min / clearance claims
Certification	Authority establishes final basis and special-condition need
Traceability	Every new claim links to a requirement, method and closure evidence
Document integrity	Rev29 retained in full; no blank/filler pages introduced by the correction package

Independent engineering review remains mandatory. The purpose of Rev33 is to make the engineering and statistical logic harder to misuse, not to declare the aircraft feasible, safe or certifiable.

REV33-23 — Handover / Reference Basis

Project sources: HSH-00 Rev29 complete baseline; Rev31 corrected baseline; Rev32 engineering-closure candidate. Rev33 is an additive correction package and does not replace the controlled Rev29 source pages in the integrated file.

Regulatory references checked for this release: EASA Easy Access Rules for Large Rotorcraft (CS-29, Amendment 12, 23 March 2026); EASA Part 21 material on Special Conditions and type-certification basis; FAA AC 29-2C rotorcraft certification guidance. These references are planning context only; they do not establish certification credit for HSH.

Handover item	State
Rev29 complete source	Preserved in integrated Rev33 PDF
Rev33 correction package	Current engineering interpretation
Independent review	Required before design use
Certification authority	Not assigned / not agreed
Flight test	None
Supplier qualification	None claimed
FEA / CFD / 6-DOF evidence	None claimed; analyses are defined as work packages
Next controlled action	Independent technical review → issue disposition → configuration-controlled Rev34 if required

The final document therefore contains a clean separation between controlled configuration, engineering screening, open evidence and future certification activity. Downstream 1000-document generation should use the Rev33 corrected interpretation as the baseline input and must stop if an upstream contradiction is found.